Ex no : 1 Basic LAN Setup and IP Addressing Between PCs

Date :

Aim

To create a basic Local Area Network (LAN) using Cisco Packet Tracer, assign IP addresses to PCs, and verify communication between them using the Ping command.

Procedure

Step 1: Open Cisco Packet Tracer

- Launch Cisco Packet Tracer and familiarize yourself with the interface.
- Network Devices Toolbar – contains routers, switches, hubs, and other devices.
- End Devices – PCs, laptops, servers, and printers.
- Connections – different categories of cables used to link devices.
- Workspace – the area used to build the network topology.

Step 2: Create the LAN Topology

- Drag and drop 1 Switch into the workspace.
- Drag and drop 1 PC(PC0) and 1 Laptop (Laptop0) into the workspace.
- Arrange the devices neatly for clarity.

Step 3: Connect the Devices

- Select Connections from the toolbar and choose the Copper Straight-Through cable.
- Connect PC0 FastEthernet0 to Switch FastEthernet0/1.
- Connect PC1 FastEthernet0 to Switch FastEthernet0/2.
- Wait until the link lights turn green, indicating successful connectivity.

Step 4 & 5: Configure IP Addresses for PC0 and PC1

- Click each PC and navigate to Desktop → IP Configuration.
- Enter the IP address and subnet mask for each PC exactly as listed in the Configuration Table below.

Step 6: Verify IP Configuration

- Open Desktop → Command Prompt on both PCs.
- Type the ipconfig command shown below.
- Verify that the assigned IP address and subnet mask are displayed correctly.

```
ipconfig
```

Step 7: Test Connectivity Using Ping

- Open the Command Prompt on PC0 and pingLaptop0.
- Observe the reply messages indicating successful communication.
- Repeat the process from Laptop0 to PC0.

```
ping 192.168.10.2
ping 192.168.10.1
```

Step 8: Observe Packet Transmission in Simulation Mode

- Switch to Simulation Mode.

- Select Add Simple PDU and click PC0, then Laptop0.
- Click Auto Capture/Play.
- Observe the ICMP Echo Request and Echo Reply packets travelling through the switch.

Step 9: View Switch Information

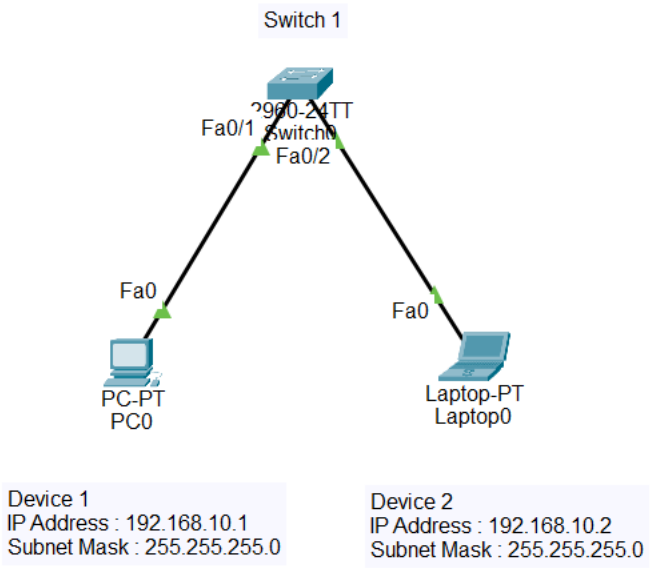
- Click the Switch and open the Config tab to view the FastEthernet interfaces, VLAN configuration, and port status.
- Open the CLI tab to observe basic switch commands.

Configuration Table

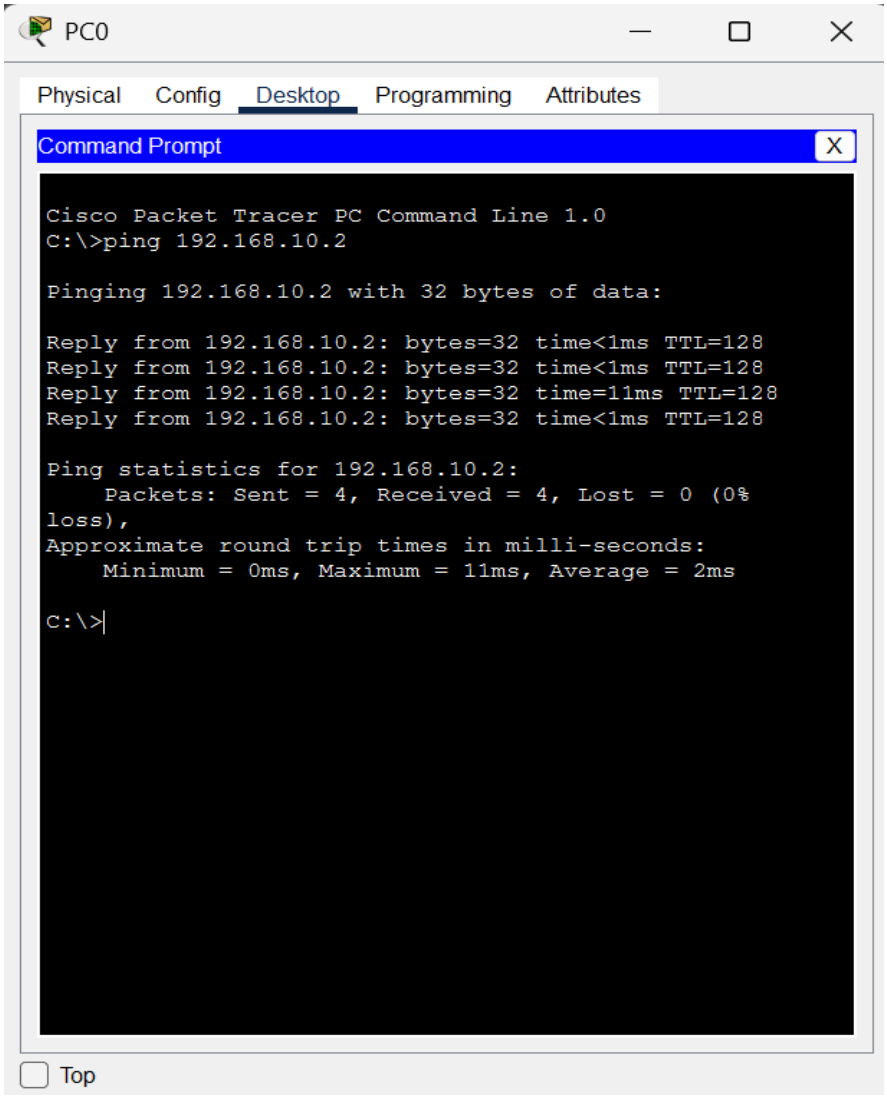
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.1	255.255.255.0	—
Laptop0	192.168.10.2	255.255.255.0	—

Circuit Diagram



Output :



Result

Thus, a basic LAN was successfully created in Cisco Packet Tracer, IP addresses were assigned to the PCs, and communication between them was verified successfully using the Ping command. The packet transmission and network operation were observed using Simulation Mode.

Ex no : 2 Network Topology Creation (Star, Bus, Ring, Mesh)

Date :

Aim

To create and study different network topologies (Star, Bus, Ring, and Mesh) using Cisco Packet Tracer and understand their structure, connectivity, and communication between devices.

Procedure

Step 1: Open Cisco Packet Tracer

- Launch Cisco Packet Tracer and review the Network Devices Toolbar, End Devices, Connections, Workspace, and the Realtime/Simulation modes.

Part A – Step 2: Create a Star Topology

- Drag and drop 1 Switch and 4 PCs (PC0–PC3).
- Place the switch at the centre with the PCs arranged around it.

Step 3: Connect the Devices (Star)

- Use the Copper Straight-Through cable.
- Connect PC0 → Switch Fa0/1, PC1 → Switch Fa0/2, PC2 → Switch Fa0/3, PC3 → Switch Fa0/4.
- Wait until all links turn green.

Step 4: Configure IP Addresses (Star)

- Assign each PC the Star-topology IP address listed in the Configuration Table below.

Step 5: Test Connectivity (Star)

- Open the Command Prompt on PC0 and verify successful communication between all PCs.

ping 192.168.1.20

Part B – Step 6: Create a Bus Topology

- Drag and drop 1 Hub and 4 PCs.
- Arrange the hub at the centre with all PCs connected to it.

Step 7: Connect the Devices (Bus)

- Use the Copper Straight-Through cable to connect each PC to the Hub.

Step 8: Configure IP Addresses (Bus)

- Assign each PC the Bus-topology IP address listed in the Configuration Table below.

Step 9: Verify Communication (Bus)

- Open the Command Prompt on any PC and ping another PC in the network.
- Observe successful packet transmission and note that, unlike the switch-based topology, all traffic on the hub is broadcast to every connected port.

Part C – Step 10: Create a Ring Topology

- Drag and drop 4 Routers (or PCs with a suitable module) and connect each device to exactly two neighbours so the cabling forms a closed loop.
- Use Copper Cross-Over cables between adjacent devices.

Step 11: Configure and Verify the Ring

- Assign IP addresses to each device on the ring as per the Configuration Table below.
- Test connectivity between non-adjacent devices and observe that traffic may traverse multiple intermediate devices to reach its destination.

Part D – Step 12: Create a Mesh Topology

- Drag and drop 4 PCs (PC0–PC3) into the workspace and arrange them so every device can be cabled directly to every other device.
- Use Copper Cross-Over cables (or a switch-free direct-connect module) to connect PC0–PC1, PC0–PC2, PC0–PC3, PC1–PC2, PC1–PC3, and PC2–PC3, so that each PC has a dedicated link to all three of the others (a full mesh).

Step 13: Configure IP Addresses (Mesh)

- Assign each PC the Mesh-topology IP address listed in the Configuration Table below.

Step 14: Verify Full-Mesh Connectivity

- Open the Command Prompt on PC0 and ping PC1, PC2, and PC3 in turn, confirming a direct link-level path to each.
- Repeat from another PC and observe that every device can reach every other device over its own dedicated link, so no single link failure can isolate a device from the rest of the network.

ping 192.168.4.20

ping 192.168.4.30

ping 192.168.4.40

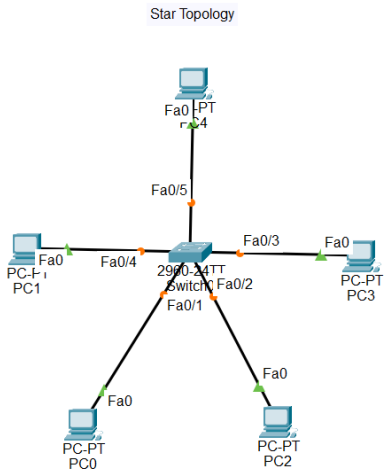
Configuration Table

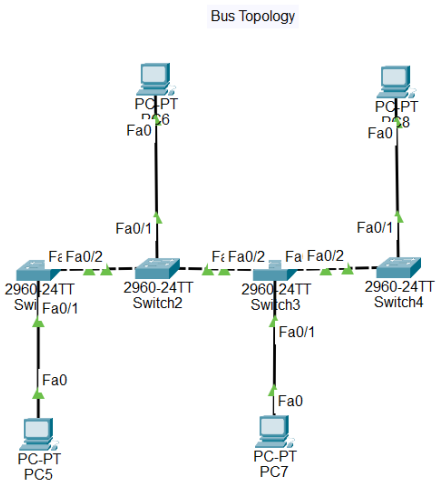
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

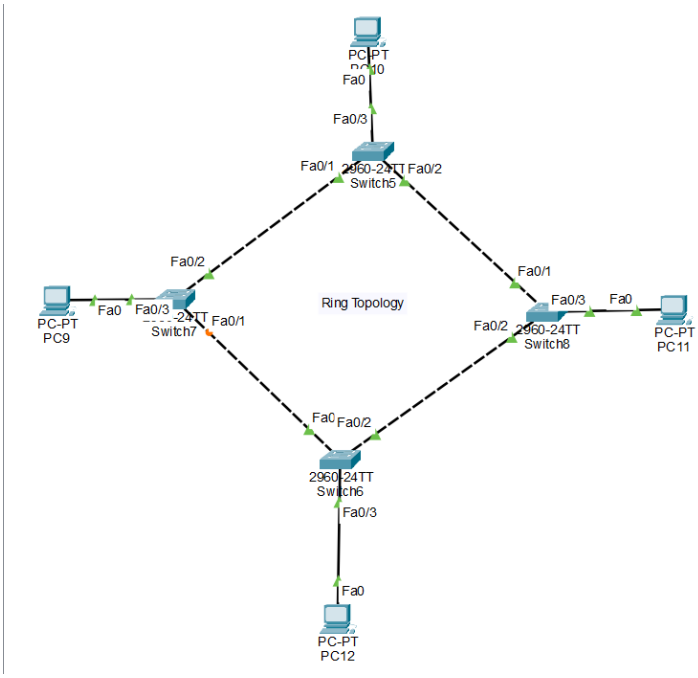
Device	IP Address	Subnet Mask	Default Gateway
PC0 (Star)	192.168.1.10	255.255.255.0	—
PC1 (Star)	192.168.1.20	255.255.255.0	—
PC2 (Star)	192.168.1.30	255.255.255.0	—
PC3 (Star)	192.168.1.40	255.255.255.0	—
PC0 (Bus)	192.168.2.10	255.255.255.0	—
PC1 (Bus)	192.168.2.20	255.255.255.0	—
PC2 (Bus)	192.168.2.30	255.255.255.0	—
PC3 (Bus)	192.168.2.40	255.255.255.0	—
Device0 (Ring)	192.168.3.10	255.255.255.0	—
Device1 (Ring)	192.168.3.20	255.255.255.0	—

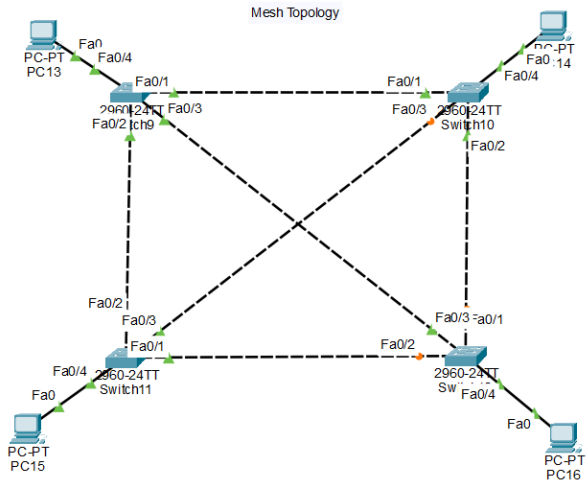
Device	IP Address	Subnet Mask	Default Gateway
Device2 (Ring)	192.168.3.30	255.255.255.0	—
Device3 (Ring)	192.168.3.40	255.255.255.0	—
PC0 (Mesh)	192.168.4.10	255.255.255.0	—
PC1 (Mesh)	192.168.4.20	255.255.255.0	—
PC2 (Mesh)	192.168.4.30	255.255.255.0	—
PC3 (Mesh)	192.168.4.40	255.255.255.0	—

Circuit Diagram

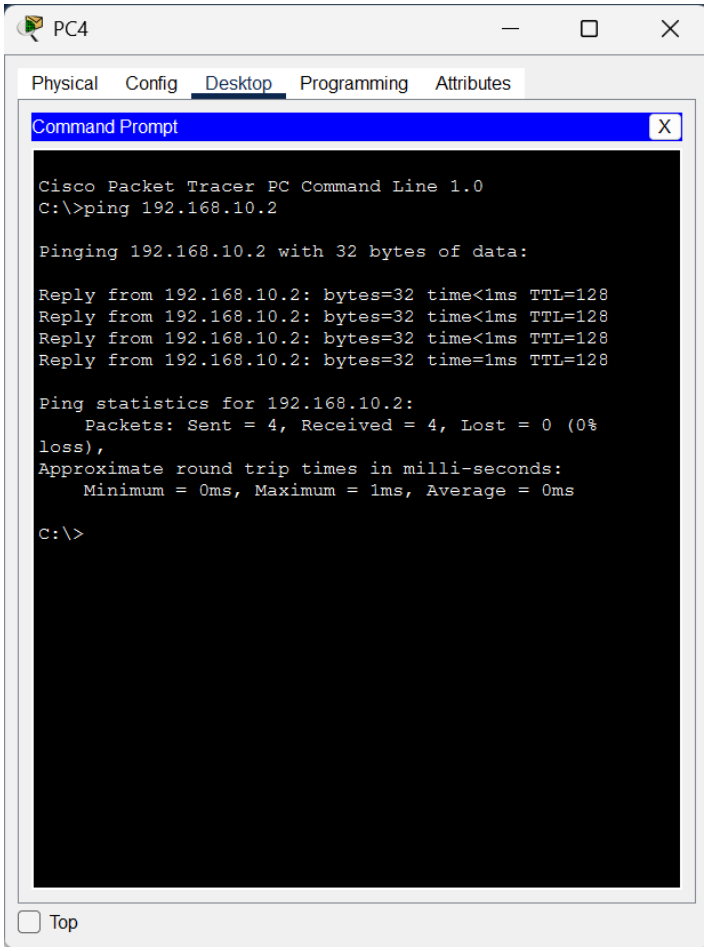





Ring Topology
PC-PT
Fa0
Fa0/3
Fa0/1
Fa0/2
Switch5
Fa0/2
Fa0/1
Fa0
Fa0/3
PC-PT
PC9
Switch7
Fa0/1
Fa0/3
Fa0
Fa0/2
Switch8
PC-PT
PC11
Fa0
Fa0/2
Switch6
Fa0/3
Fa0
PC-PT
PC12


Mesh Topology
PC-PT
PC13
Fa0
Fa0/4
Fa0/1
Fa0/3
Fa0/1
Fa0/3
Fa0/4
Fa0/2
Switch10
Fa0/2
Fa0/3
Fa0/1
Fa0/3
Fa0/1
Fa0/4
Switch11
Fa0/2
Fa0/4
Fa0
Fa0
PC-PT
PC15
PC-PT
PC16

Output :



Result

Thus, Star, Bus, Ring, and Mesh network topologies were successfully created and studied in Cisco Packet Tracer. The structure and cabling pattern of each topology were examined, and communication between devices was verified in each case, highlighting the differences in traffic handling and redundancy between switch-based, hub-based, ring-based, and fully meshed connectivity.

Ex no : 3 Ping Command for Connectivity Testing

Date :

Aim

To use the Ping command in Cisco Packet Tracer to test connectivity between network devices and to interpret the results obtained, including round-trip time, TTL, and packet loss.

Procedure

Step 1: Build the Test Topology

- Drag and drop 1 Switch ,2 PCs (PC0, PC1) and one Server into the workspace.
- Connect each PC to the switch using Copper Straight-Through cables.
- Wait for all link lights to turn green.

Step 2: Assign IP Addresses

- Configure each PC with the IP address listed in the Configuration Table below.

Step 3: Perform a Basic Ping Test

- Open the Command Prompt on PC0.
- Ping PC1 and Server0 and observe the four Echo Reply messages, along with the round-trip time (RTT) and Time-To-Live (TTL) values.

  ping 192.168.10.2 , ping 192.168.10.3

Step 4: Perform an Extended (Continuous) Ping

- Use the -t option to send continuous ping requests until manually stopped.
- Press Ctrl+C to stop the continuous ping and view the summary statistics.

  ping -t 192.168.10.3

Step 5: Test an Unreachable Host

- Ping an IP address that does not exist on the network, such as 192.168.10.50.
- Observe the Request Timed Out messages, which indicate packet loss or an unreachable destination.

  ping 192.168.10.50

Step 6: Trace the Route

- Use the tracert command to view the path taken by packets to reach the destination and the number of hops involved.

  tracert 192.168.10.3

Step 7: Observe in Simulation Mode

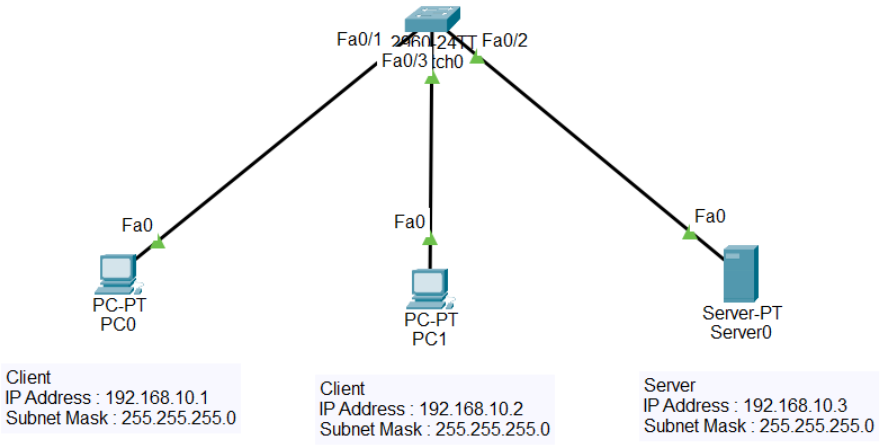
- Switch to Simulation Mode, add a Simple PDU between PC0 and PC2, and step through the event list.
- Examine the ICMP Echo Request and Echo Reply packets at each layer using the PDU information window.

Configuration Table

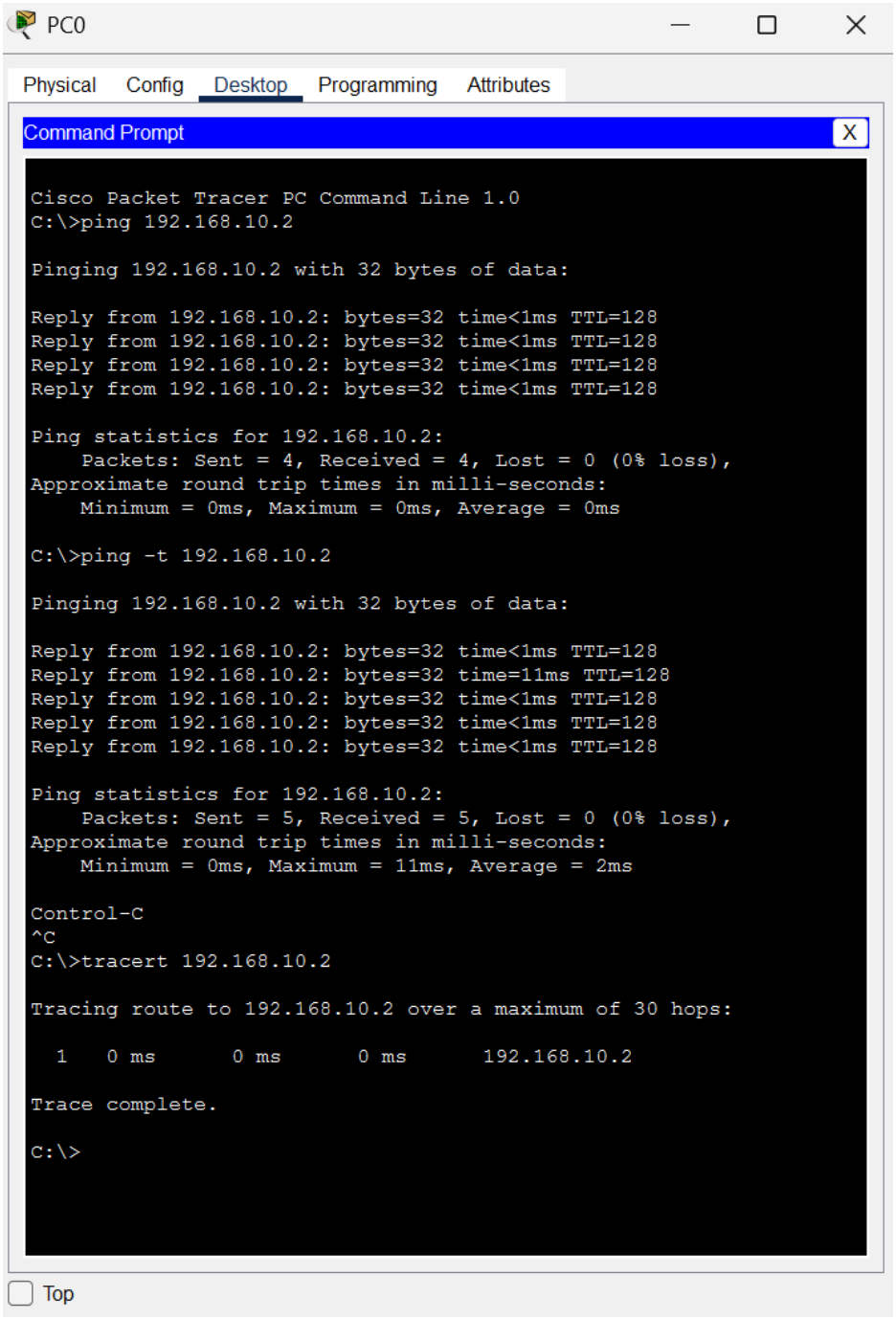
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.1	255.255.255.0	—
PC1	192.168.10.2	255.255.255.0	—
Server 0	192.168.10.3	255.255.255.0	—

Circuit Diagram



Output :



Result

Thus, the Ping command was used successfully to test and verify connectivity between devices in Cisco Packet Tracer. Reachable and unreachable hosts were tested, round-trip time and TTL values were interpreted, and the path of packets was traced using the tracert command.

Ex no : 4 ARP Protocol Simulation in LAN

Date :

Aim

To simulate and study the working of the Address Resolution Protocol (ARP) in a LAN using Cisco Packet Tracer, and to observe how IP addresses are resolved to MAC addresses.

Procedure

Step 1: Build the Topology

- Drag and drop 1 Switch and 4 PCs (PC0, PC1, PC2 and PC3).
- Connect PC0 and PC1 to the switch using Copper Straight-Through cables.

Step 2: Configure IP Addresses

- Configure PC0 and PC1 with the IP addresses listed in the Configuration Table below.

Step 3: View and Clear the ARP Cache

- Open the Command Prompt on PC0 and display the current ARP cache using the arp -a command.
- Clear the cache so that a fresh ARP exchange can be observed in the next step.

```
arp -a
arp -d
```

Step 4: Generate ARP Traffic

- Switch to Simulation Mode and use Edit Filters to select only ARP and ICMP protocols.
- Select Add Simple PDU and click PC0, then PC1, to send a ping request.
- Click Auto Capture/Play to run the simulation.

Step 5: Observe the ARP Request

- Note that PC0 first broadcasts an ARP Request frame to the entire LAN.
- Click on the ARP packet event in the event list and inspect the PDU details at the Network layer.

Step 6: Observe the ARP Reply

- Observe that PC1 responds with a unicast ARP Reply frame containing its MAC address.
- Note that only after the ARP exchange completes does PC0 send the actual ICMP Echo Request to PC1.

Step 7: Verify the Updated ARP Cache

- Return to Realtime Mode and display the ARP cache on PC0 again.
- Confirm that an entry now maps PC1's IP address to its MAC address.

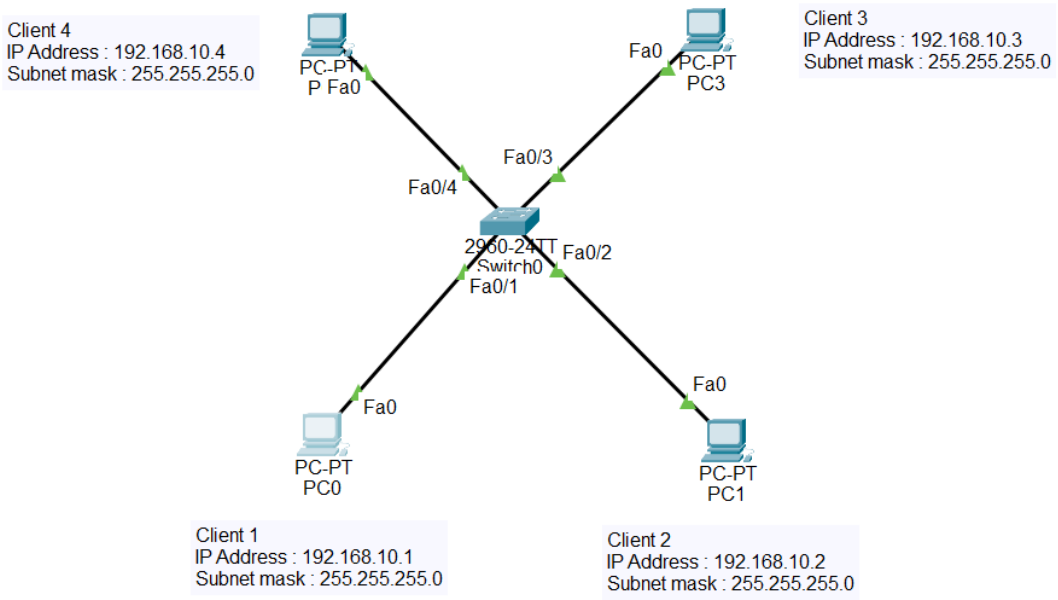
```
arp -a
```

Configuration Table

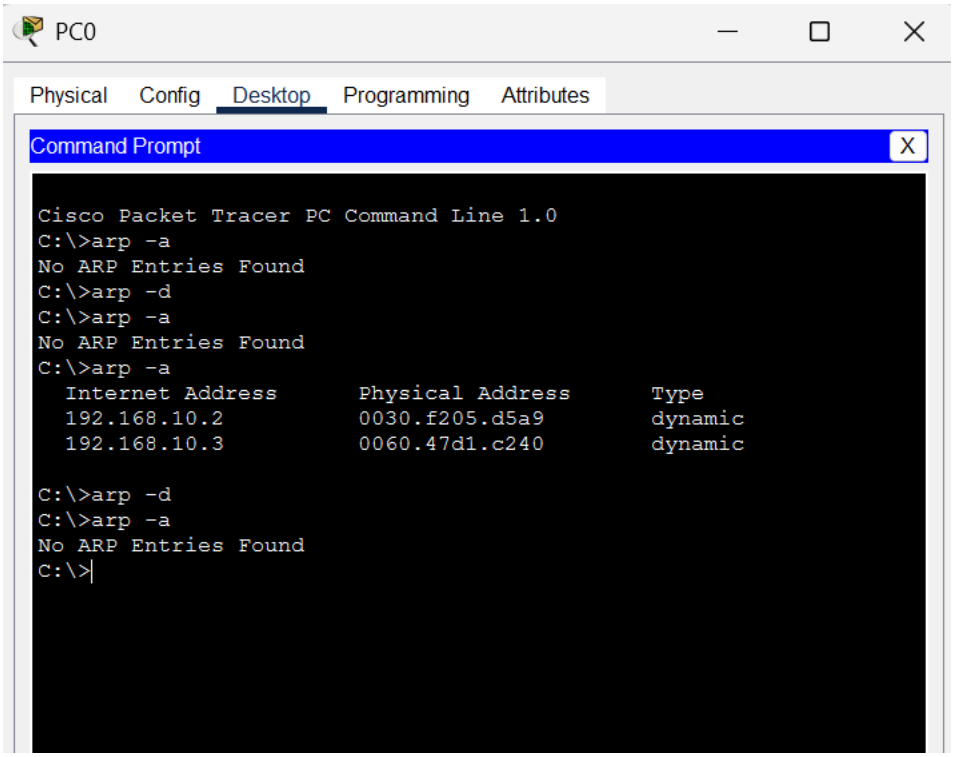
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.20.1	255.255.255.0	—
PC1	192.168.20.2	255.255.255.0	—
PC2	192.168.20.3	255.255.255.0	—
PC4	192.168.20.4	255.255.255.0	—

Circuit Diagram



Output :



Result

Thus, the working of the ARP protocol was successfully simulated in Cisco Packet Tracer. The broadcast ARP Request and unicast ARP Reply were observed, and the resolution of an IP address to its corresponding MAC address was verified using the ARP cache.

Ex no : 5 Ethernet LAN Setup Using Switch

Date :

Aim

To configure an Ethernet LAN using a Layer 2 switch in Cisco Packet Tracer and to examine how the switch learns and maintains MAC addresses in its address table.

Procedure

Step 1: Build the Topology

- Drag and drop 1 Switch and 1 PC (PC0) and 1 Laptop (Laptop0) .
- Connect each PC to a separate FastEthernet port on the switch using Copper Straight-Through cables.

Step 2: Configure IP Addresses

- Configure each PC with the IP address listed in the Configuration Table below.

Step 3: Assign a Display Name to the Switch

- Click the switch to open its configuration window and select the Config tab.
- Under Global Settings, enter LAN_SW1 in the Display Name field to identify the switch.

Step 4: Verify Interface Settings

- Still on the Config tab, select each FastEthernet interface in turn.
- Confirm that Port Status is set to On and note the Bandwidth and Duplex fields for each port.

Step 5: Generate Traffic Between PCs

- From PC0, ping PC1, PC2, and PC3 in turn so that the switch has traffic to learn from.

```
ping 192.168.10.12
ping 192.168.10.13
```

Step 6: View the MAC Address Table (CLI)

- Packet Tracer does not provide a graphical view of the dynamically learned MAC address table, so open the switch's CLI tab only for this read-only check.
- Display the MAC address table and verify that each PC's MAC address is associated with the correct FastEthernet port.

```
Switch>enable
Switch#show mac-address-table
```

Step 7: Verify Port Status

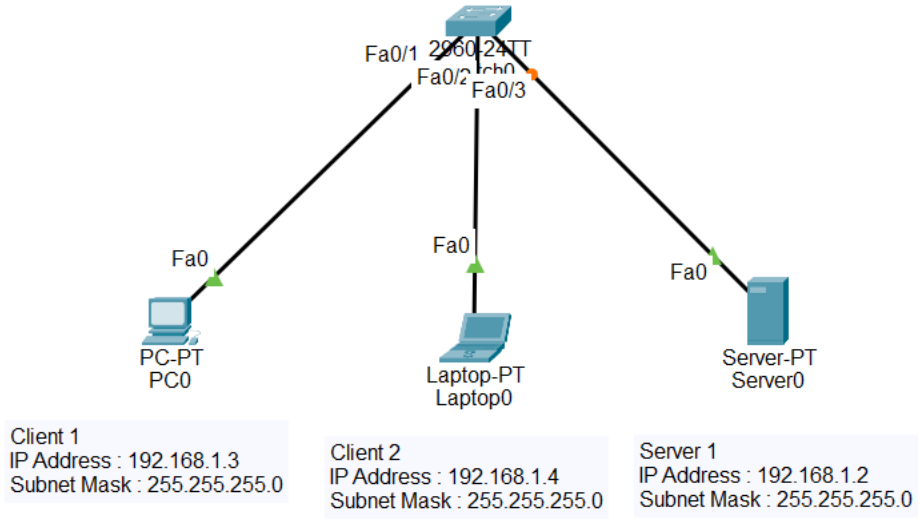
- Return to the Config tab on the switch and check the status and speed/duplex settings of each FastEthernet interface once more.

Configuration Table

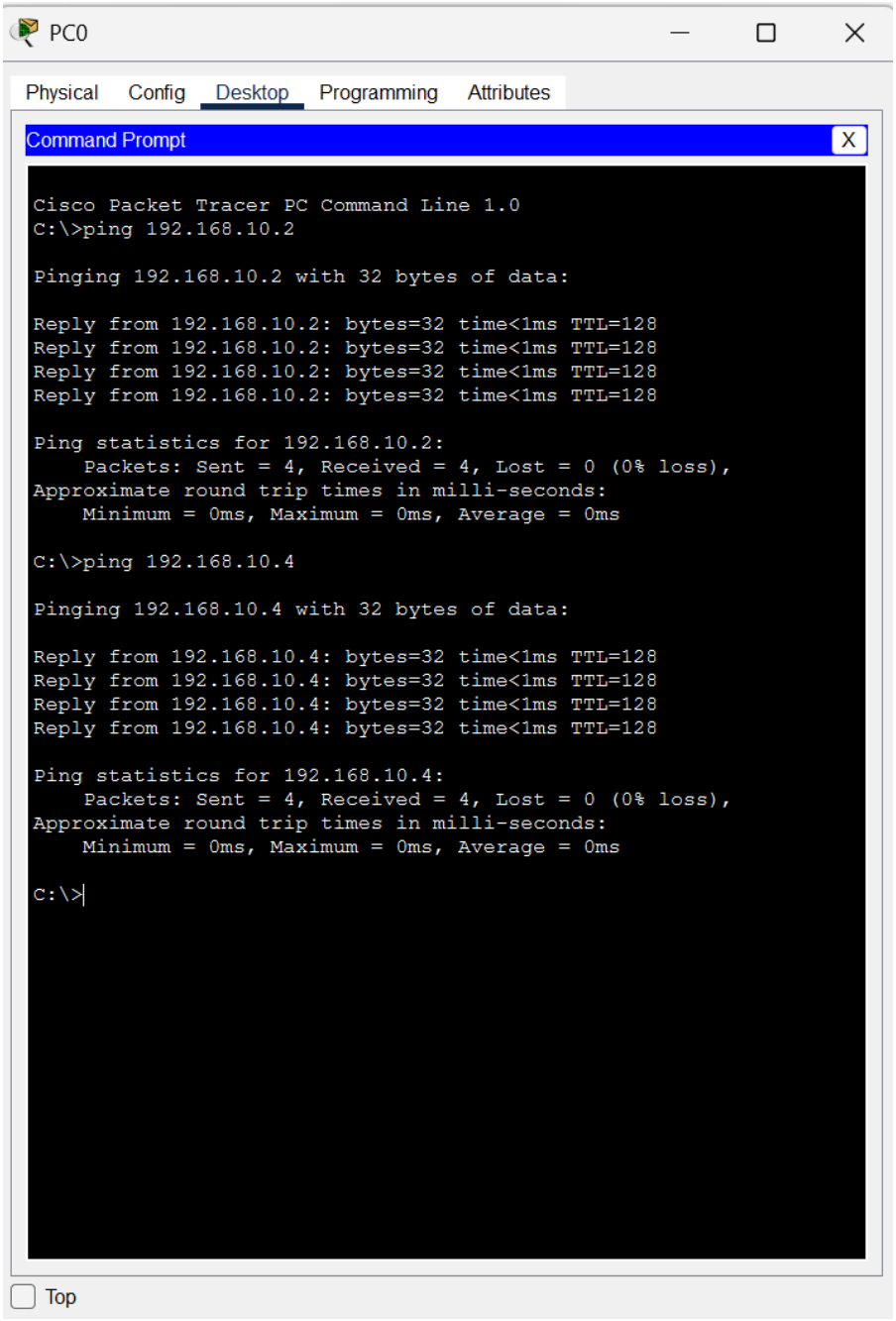
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

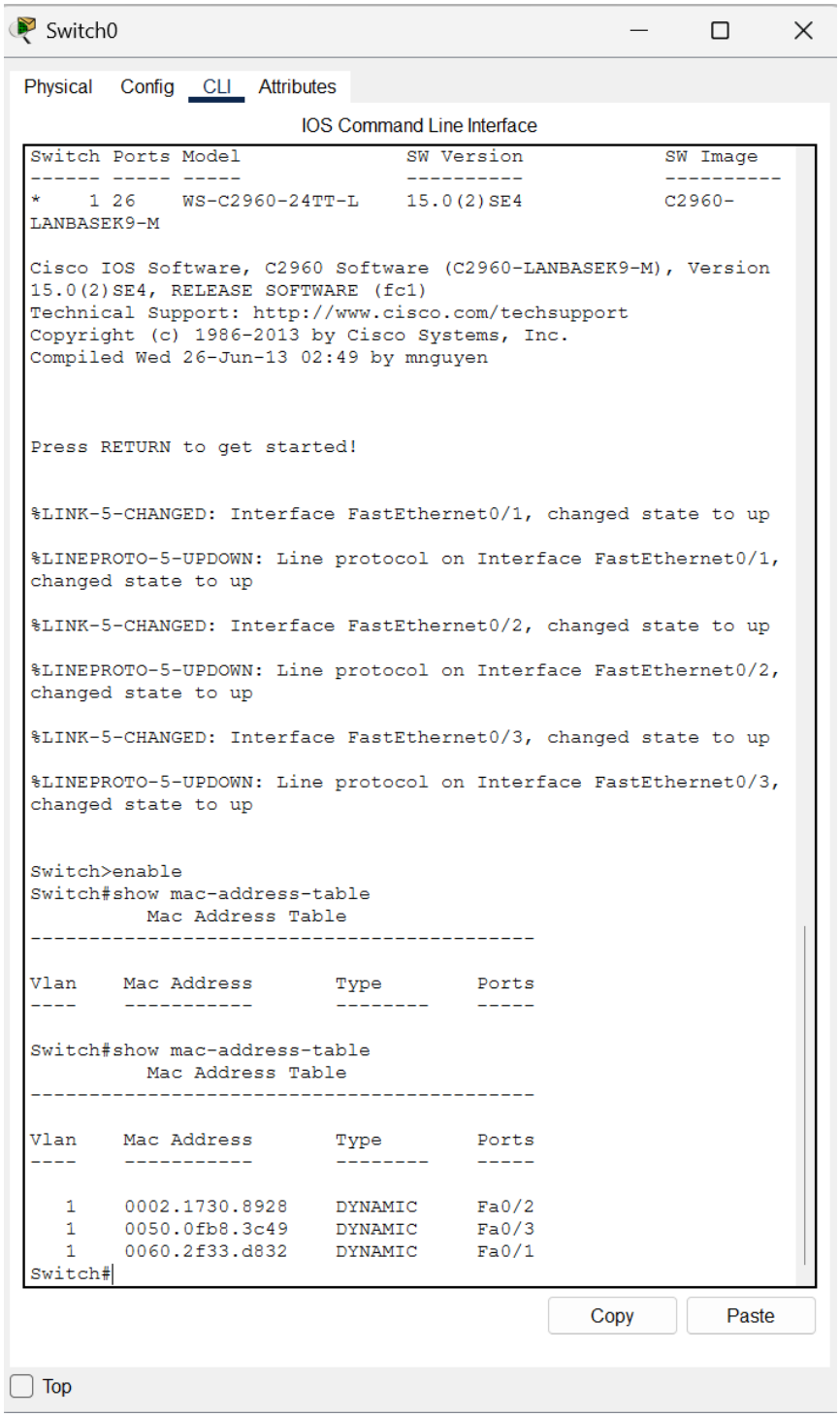
Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.3	255.255.255.0	—
Laptop0	192.168.10.4	255.255.255.0	—
Server0	192.168.10.2	255.255.255.0	—
Switch0 (LAN_SW1)	Not configured	—	—

Circuit Diagram



Output :


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2
Pinging 192.168.10.2 with 32 bytes of data:
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.10.4
Pinging 192.168.10.4 with 32 bytes of data:
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top



Result

Thus, an Ethernet LAN was successfully configured using a switch in Cisco Packet Tracer. Connectivity between all PCs was verified, and the switch's dynamically learned MAC address table was examined to confirm correct Layer 2 operation.

Ex no : 6 Wireless LAN (Wi-Fi) Configuration

Date :

Aim

To configure a wireless LAN using an Access Point in Cisco Packet Tracer, connect wireless end devices to it, and verify connectivity between them.

Procedure

Step 1: Build the Topology

- Drag and drop 1 Access Point (Access Point-PT) into the workspace.
- Drag and drop 4 End devices into the workspace.

Step 2: Add Wireless Interfaces

- Click each laptop, go to the Physical tab, power it off, remove the wired NIC module, and insert a WPC300N wireless module, then power it back on.

Step 3: Configure the Access Point

- Click the Access Point and open the Config tab.
- Select the Port1 (wireless) interface and set the Network Mode, Channel, and SSID (for example, LabWiFi).

Step 4: Configure Wireless Security

- In the Access Point's GUI, choose WPA2-PSK authentication and set a pre-shared key (passphrase).
- On each laptop's Desktop → PC Wireless application, select the Connect tab, choose the SSID LabWiFi, and enter the same passphrase.

Step 5: Configure IP Addresses

- Open Desktop → IP Configuration on each laptop and enter the IP details listed in the Configuration Table below.

Step 6: Verify the Wireless Link

- Confirm that the link lights between each laptop and the Access Point are green, indicating an associated wireless connection.
- Open PC Wireless on each laptop and check the signal strength on the Connect tab.

Step 7: Test Connectivity

- Open the Command Prompt on Laptop0 and ping PC0 to confirm successful wireless communication.

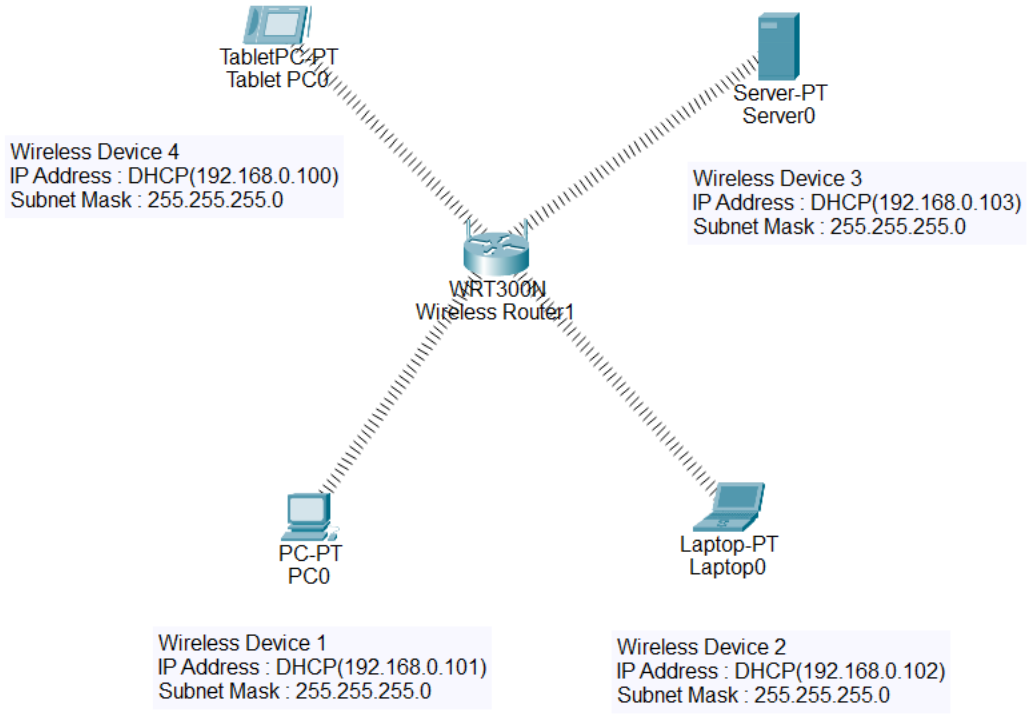
  ping 192.168.10.2

Configuration Table

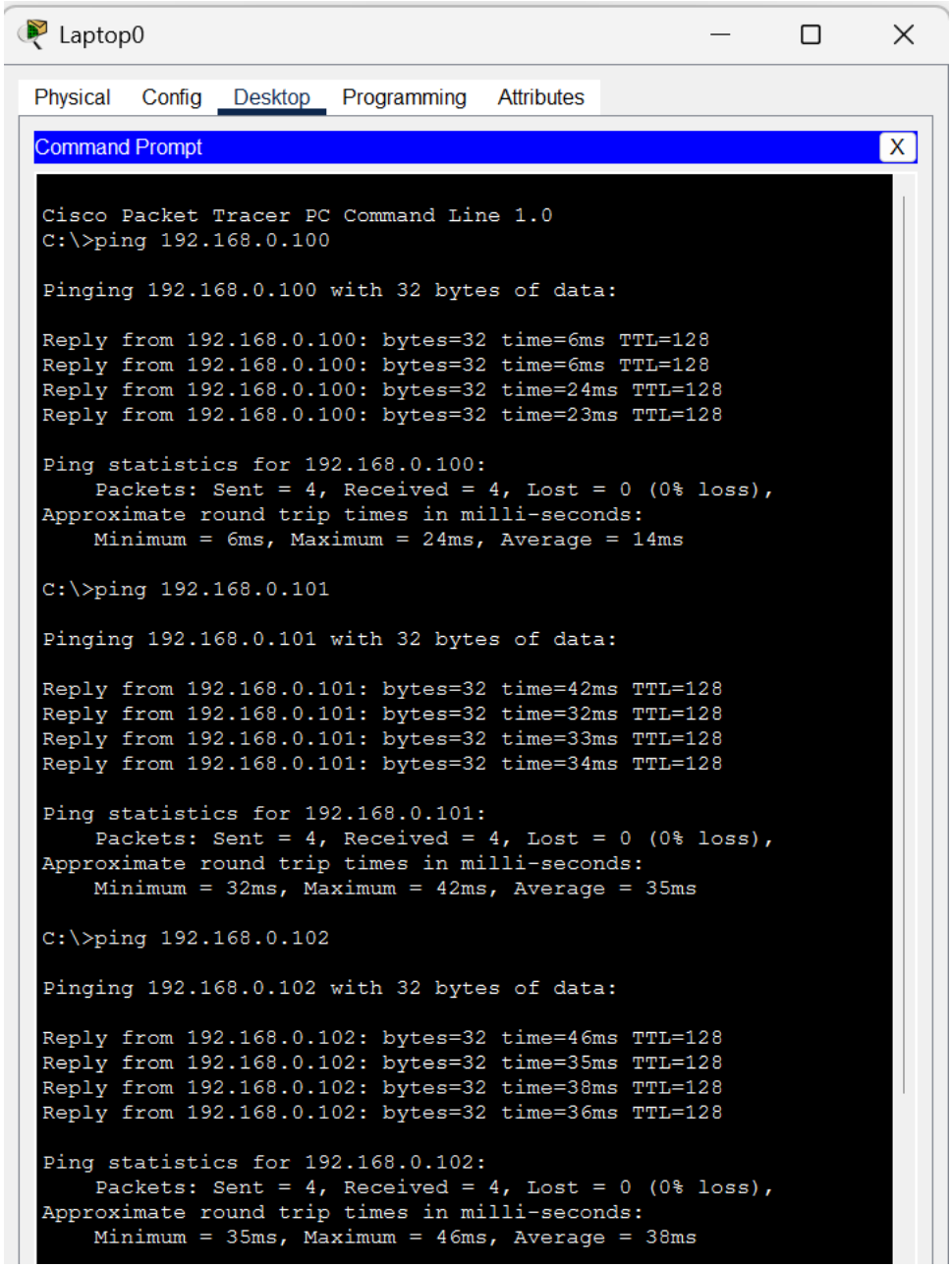
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	Interface	IP Address(DHCP)	Subnet Mask	Default Gateway
PC0	WPC300N (wireless)	192.168.0.101	255.255.255.0	—
Laptop0	WPC300N (wireless)	192.168.0.102	255.255.255.0	—
Server0	WPC300N (wireless)	192.168.0.103	255.255.255.0	—
TabletPC0	WPC300N (wireless)	192.168.0.100	255.255.255.0	—
AccessPoint0	Port1 (SSID: LabWiFi)	Not configured	—	—

Circuit Diagram



Output :



Result

Thus, a Wireless LAN was successfully configured in Cisco Packet Tracer using an Access Point with WPA2-PSK security. Wireless devices were associated with the Access Point, IP addresses were assigned, and connectivity between the wireless laptops was verified.

Ex no : 7 Static Routing Between Two Networks

Date :

Aim

To configure static routing between two separate networks using Cisco routers in Cisco Packet Tracer and verify inter-network communication.

Procedure

Step 1: Build the Topology

- Drag and drop 2 Routers (2911), 2 Switches (2960), and 4 PCs (PC0 – PC3).
- Connect PC0,PC1 → Switch0 → Router0 (GigabitEthernet0/0) and PC2,PC3 → Switch1 → Router1 (GigabitEthernet0/0).
- Connect Router0 (GigabitEthernet0/1) to Router1 (GigabitEthernet0/1) using a Copper Cross-Over cable.

Step 2: Plan the IP Addressing Scheme

- Refer to the Configuration Table below for the addressing plan across both routers and PCs.

Step 3: Configure Router0 Interfaces

- Click Router0 and select the Config tab.
- Select the GigabitEthernet0/0 interface, enter its IP Address and Subnet Mask from the Configuration Table, then set Port Status to On.
- Select the GigabitEthernet0/1 interface, enter its IP Address and Subnet Mask from the Configuration Table, then set Port Status to On.

Step 4: Configure Router1 Interfaces

- Click Router1 and select the Config tab.
- Select the GigabitEthernet0/0 interface, enter its IP Address and Subnet Mask from the Configuration Table, then set Port Status to On.
- Select the GigabitEthernet0/1 interface, enter its IP Address and Subnet Mask from the Configuration Table, then set Port Status to On.

Step 5: Configure Static Routes

- On Router0's Config tab, open ROUTING → STATIC.
- Enter Network 192.168.2.0, Mask 255.255.255.0, and Next Hop IP 10.0.0.2, then click Add.
- On Router1's Config tab, open ROUTING → STATIC.
- Enter Network 192.168.1.0, Mask 255.255.255.0, and Next Hop IP 10.0.0.1, then click Add.

Step 6: Configure PC IP Addresses and Gateways

- Set PC0 and PC1's IP address, subnet mask, and default gateway as listed in the Configuration Table below.

Step 7: Verify the Static Routes

- Return to the ROUTING → STATIC page on each router and confirm that the entered route appears in the list.

Step 8: Test End-to-End Connectivity

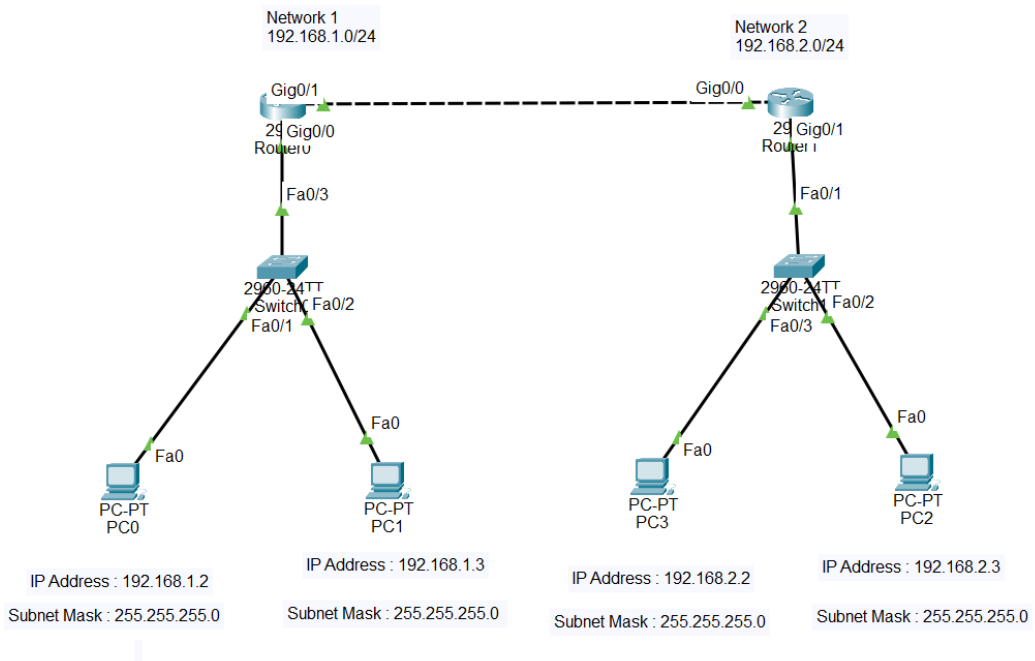
- From PC0, ping PC1 and confirm successful replies across both networks.
  ping 192.168.2.10

Configuration Table

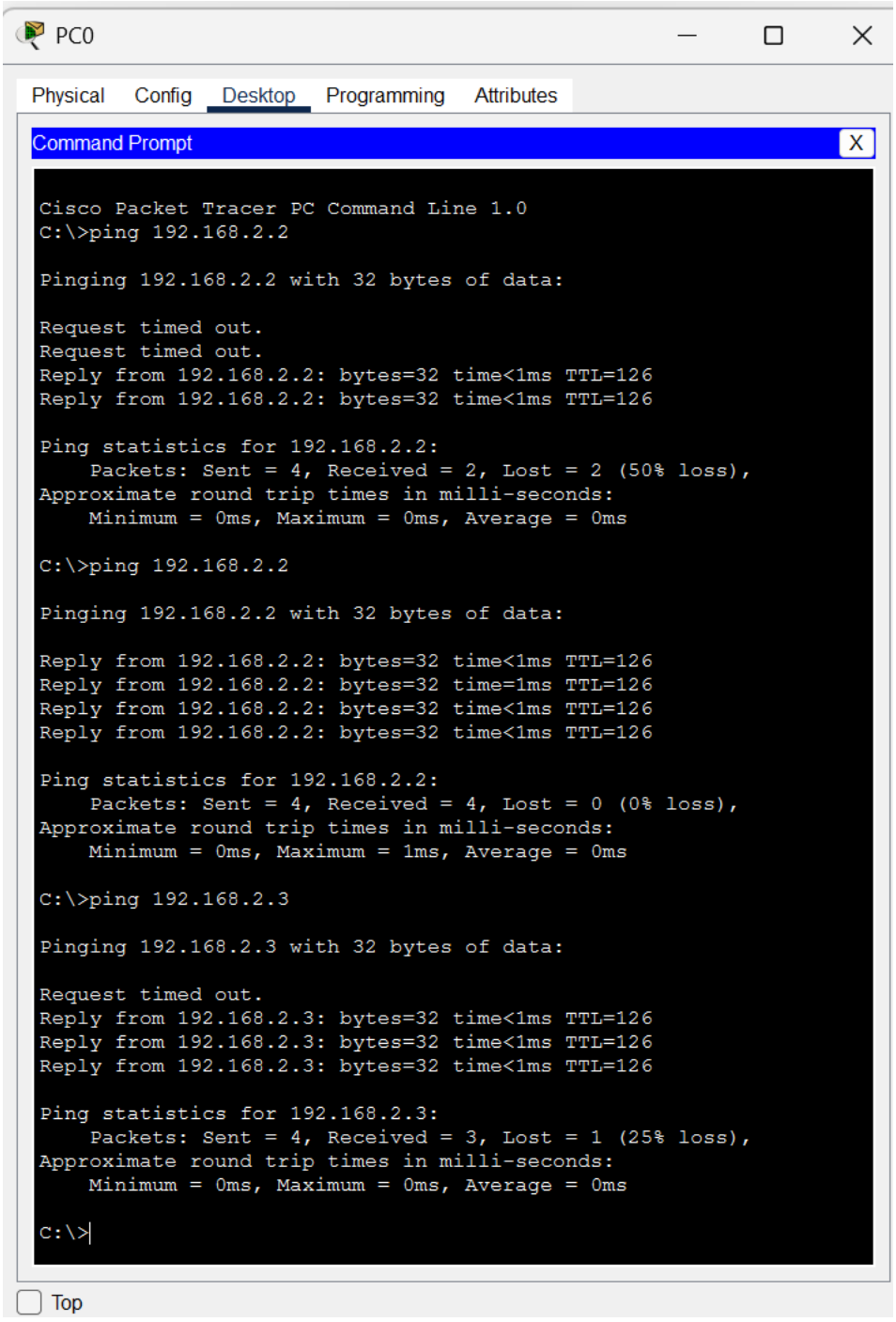
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.3	255.255.255.0	192.168.1.1
Router0	Gi0/0	192.168.1.1	255.255.255.0	—
Router0	Gi0/1	10.0.0.1	255.255.255.252	—
Router1	Gi0/1	10.0.0.2	255.255.255.252	—
Router1	Gi0/0	192.168.2.1	255.255.255.0	—
PC2	NIC	192.168.2.2	255.255.255.0	192.168.2.1
PC3	NIC	192.168.2.3	255.255.255.0	192.168.2.1

Circuit Diagram



Output :



Result

Thus, static routing was successfully configured between two networks using Cisco routers in Packet Tracer. The routing tables were verified, and end-to-end connectivity between PC0 and PC1 across the two networks was confirmed.

Ex no : 8 RIP Routing Configuration (Basic Dynamic Routing)

Date :

Aim

To configure the Routing Information Protocol (RIP) on Cisco routers in Packet Tracer to enable dynamic exchange of routing information between networks.

Procedure

Step 1: Build the Topology

- Drag and drop 3 Routers 3 Switches and 6 PCs (PC0-5).
- Connect PC0,PC1 → Switch0 → Router0, PC2,PC3 → Switch1 → Router1 and PC4,PC5 → Switch2 → Router2.
- Connect Router0 ↔ Router1 and Router1 ↔ Router2 using Copper Cross-Over cables, forming a three-router chain.

Step 2: Plan the IP Addressing Scheme

- Refer to the Configuration Table below for the full addressing plan.
- Click each router, select the Config tab, and select each of its interfaces in turn.
- Enter the IP Address and Subnet Mask from the Configuration Table for that interface, then set Port Status to On.
- Repeat for both interfaces on Router0, both interfaces on Router1, and both interfaces on Router2.

Step 4: Enable RIP on Router0

- On Router0's Config tab, open ROUTING → RIP.
- Enter Network 192.168.10.0 and click Add.
- Enter Network 10.10.10.1 and click Add.

Step 5: Enable RIP on Router1 and Router2

- On Router1's Config tab, open ROUTING → RIP and add both of its connected networks: 10.10.10.2, 10.10.20.1 and 192.168.20.0 .
- On Router2's Config tab, open ROUTING → RIP and add both of its connected networks: 10.10.20.2 and 192.168.30.0.

Step 6: Allow Time for RIP to Converge

- Packet Tracer's GUI does not display the computed routing table, so RIP operation is confirmed practically through the connectivity test in the next step.
- Wait a few seconds after adding the networks to allow the routers to exchange routing updates.

Step 7: Test End-to-End Connectivity

- Configure PC0 and PC1 with the IP addresses and default gateways listed in the Configuration Table.
- Ping from PC0 to PC1 and confirm that packets are successfully routed across all three routers.

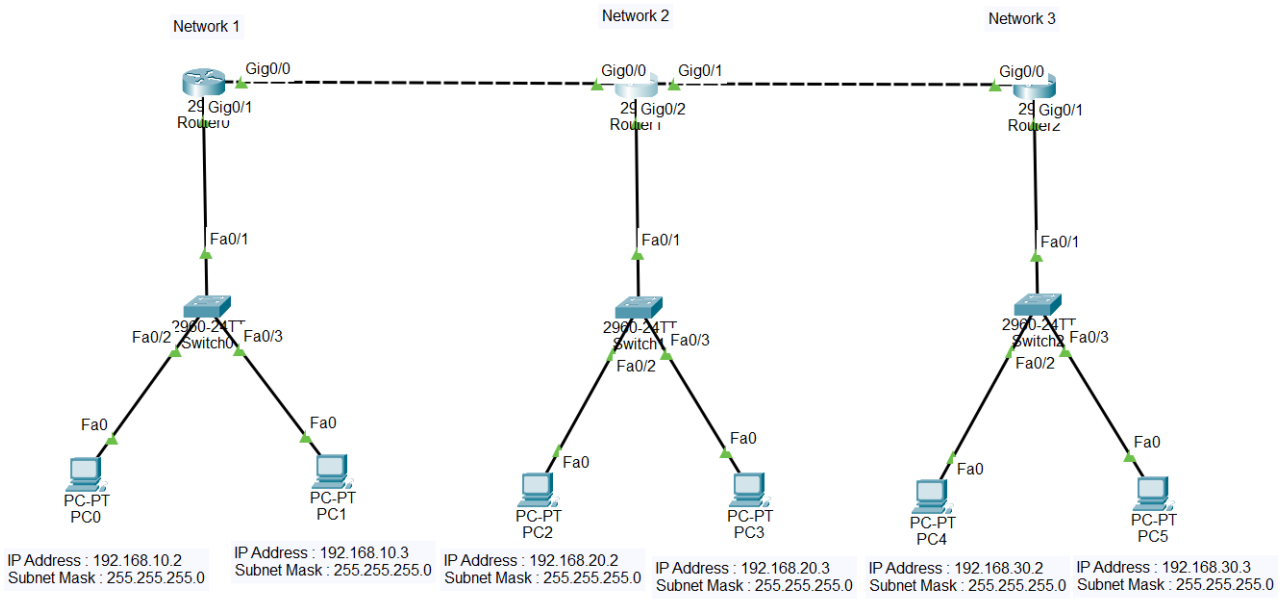
ping 192.168.20.1

Configuration Table

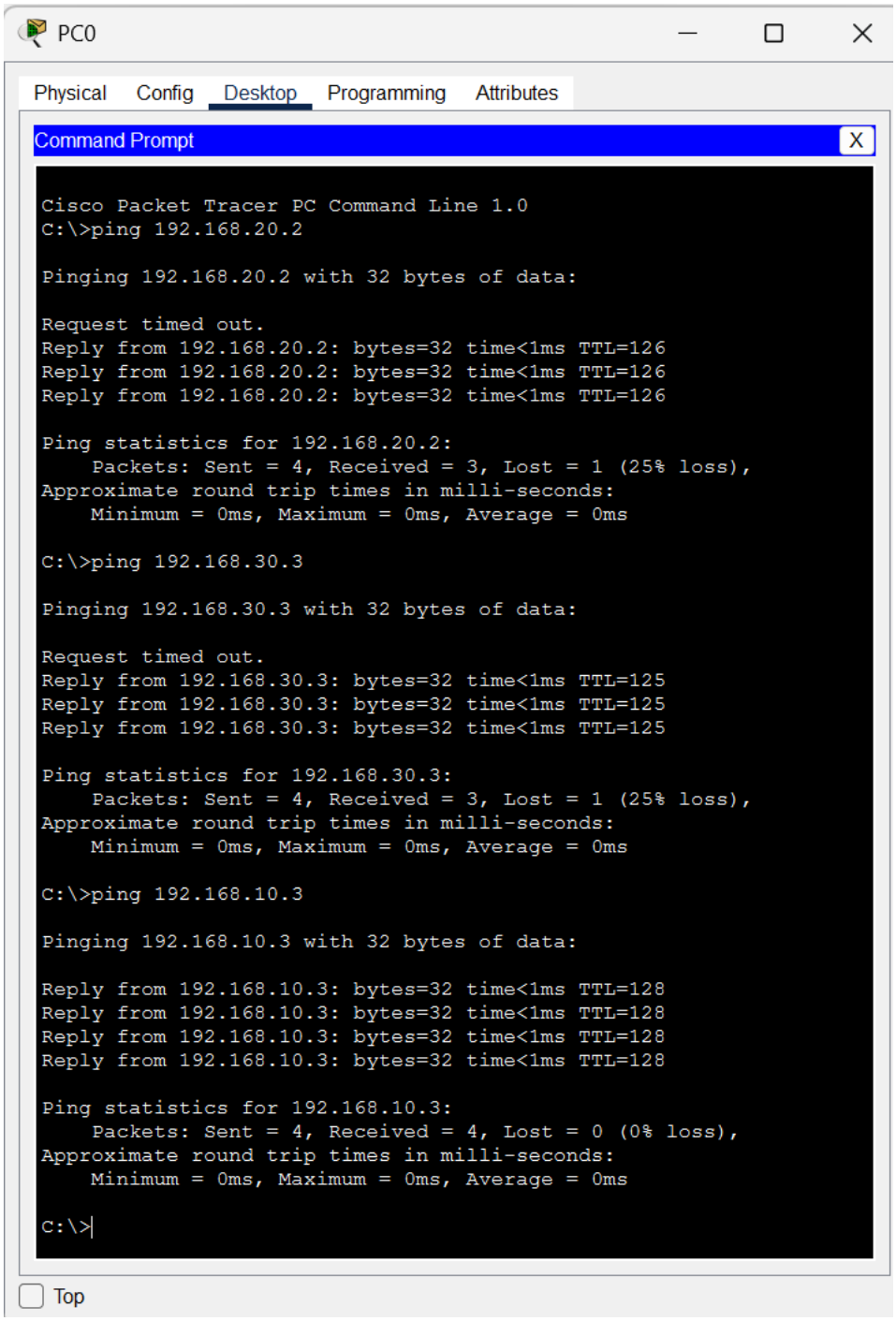
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	NIC	192.168.10.2	255.255.255.0	192.168.10.1
PC1	NIC	192.168.10.3	255.255.255.0	192.168.10.1
Router0	LAN interface	192.168.10.1	255.255.255.0	—
Router0	WAN link 1	10.0.10.1	255.255.255.252	—
Router1	WAN link 1	10.0.10.2	255.255.255.252	—
Router1	WAN link 2	10.0.20.1	255.255.255.252	—
Router1	WAN link 3	192.168.20.1	255.255.255.0	—
PC2	NIC	192.168.20.2	255.255.255.0	192.168.20.1
PC3	NIC	192.168.20.3	255.255.255.0	192.168.20.1
Router2	WAN link 2	10.10.20.2	255.255.255.252	—
Router2	LAN interface	192.168.30.1	255.255.255.0	—
PC4	NIC	192.168.30.2	255.255.255.0	192.168.30.1
PC5	NIC	192.168.30.3	255.255.255.0	192.168.30.1

Circuit Diagram



Output :



Result

Thus, RIP was successfully configured on the routers in Cisco Packet Tracer. Dynamic exchange of routing information was verified through the routing tables, and end-to-end connectivity between the two remote LANs was confirmed.

Ex no : 9 VLAN Creation and Simple Switch Configuration

Date :

Aim

To create Virtual LANs (VLANs) on a Cisco switch in Packet Tracer, assign switch ports to the VLANs, and verify that hosts in different VLANs are isolated at Layer 2.

Procedure

Step 1: Build the Topology

- Drag and drop 1 Switch and 6 PCs (PC0–PC5).
- Connect PC0 to port Fa0/1 , PC1 to port Fa0/2 and PC2 to port Fa0/3.

Step 2: Plan the VLAN and IP Scheme

- Refer to the Configuration Table below for the VLAN membership, switch port, and IP address of every PC.

Step 3: Create the VLANs

- Click the switch, select the Config tab, and open VLAN Database.
- Enter VLAN Number 10 and VLAN Name Sales, then click Add.
- Enter VLAN Number 20 and VLAN Name HR, then click Add.

Step 4: Assign Ports to VLANs

- Still on the Config tab, select interface FastEthernet0/1.
- Confirm the port's Trunking Mode is set to Access, then choose VLAN 10 from the VLAN drop-down list.
- Repeat for FastEthernet0/2, selecting VLAN 10.
- Repeat for FastEthernet0/3 and FastEthernet0/4, selecting VLAN 20 for both.

Step 5: Verify VLAN Configuration

- Return to the VLAN Database page and confirm that VLAN 10 , VLAN 20 are listed.
- Reopen each FastEthernet interface and confirm that the correct VLAN is selected in the VLAN drop-down list.

Step 6: Test Connectivity Within and Across VLANs

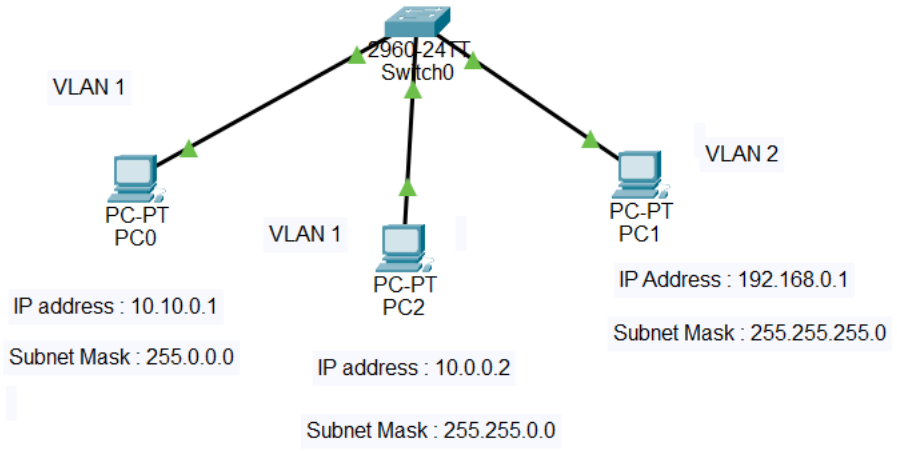
- Configure the PCs with the IP addresses from the Configuration Table above.
- Ping PC1 from PC0 and confirm success.
- Ping PC2 from PC1 and confirm failure as we haven’t created inter VLAN routing.

Configuration Table

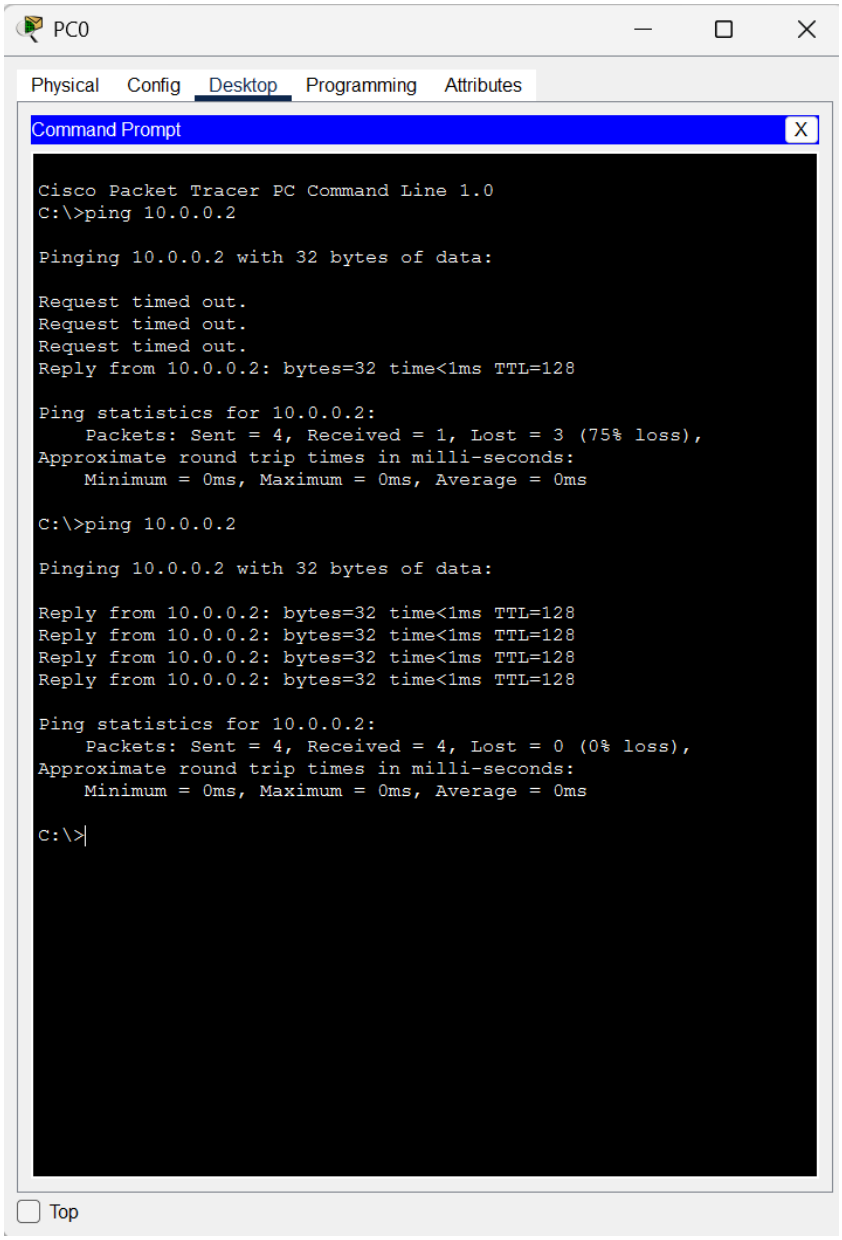
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	Fa0/0 — VLAN 10 (Sales)	10.0.0.1	255.255.0.0	—
PC1	Fa0/1 — VLAN 10 (Sales)	10.0.0.2	255.255.0.0	—
PC2	Fa0/2 — VLAN 20 (HR)	192.168.0.1	255.255.255.0	—
Switch0	None	None	None	None

Circuit Diagram



Output :



Result

Thus, VLANs were successfully created and configured on the switch in Cisco Packet Tracer. Ports were assigned to the appropriate VLANs, and it was verified that devices within the same VLAN could communicate while devices in different VLANs remained isolated.

Ex no : 10 Inter-VLAN Communication Setup

Date :

Aim

To configure inter-VLAN routing using the router-on-a-stick method in Cisco Packet Tracer .

Procedure

Step 1: Build the Topology

- Drag and drop 1 Router (2911), 3 Switches, and 4 PCs (PC0–PC3).
- Connect the comments and configure them as per the configuration table given below.
- Connect the switch's GigabitEthernet0/0 port to the router's GigabitEthernet0/0 port using a Copper Straight-Through cable — this single link will carry traffic for both VLANs.

Step 2: Create the VLANs on the Switch (GUI)

- Click the switch, select the Config tab, and open VLAN Database.
- Enter VLAN Number 10 and VLAN Name Sales, then click Add.
- Enter VLAN Number 20 and VLAN Name HR, then click Add.

Step 3: Assign Access Ports to the VLANs (GUI)

- Select interface FastEthernet0/1, confirm Trunking Mode is Access, and choose VLAN 10 from the VLAN drop-down list.
- Repeat for FastEthernet0/2, selecting VLAN 10.
- Repeat for FastEthernet0/3 and FastEthernet0/4, selecting VLAN 20 for both.

Step 4: Configure the Trunk Port Toward the Router (GUI)

- Select the GigabitEthernet0/1 interface (the port connected to the router).
- Set its Trunking Mode to Trunk so that it can carry tagged traffic for both VLAN 10 and VLAN 20 to the router on a single link.

Step 5: Configure the Router Sub-Interfaces (Router CLI — required)

- Packet Tracer's router Config (GUI) tab does not provide any option to create sub-interfaces or set 802.1Q encapsulation, so this step can only be performed from the router's CLI tab.

- Click the router, open the CLI tab, and enter global configuration mode.

```
Router>enable
Router#configure terminal
Router(config)#interface gigabitEthernet0/0.10
```

```
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#end
```

Step 6: Configure the PCs (GUI)

- Assign each PC the IP address, subnet mask, and default gateway listed in the Configuration Table below — the default gateway for each PC is its VLAN's router sub-interface address.

Step 7: Verify the Router Sub-Interfaces (Router CLI)

- From the router CLI, display the interface summary and confirm that both sub-interfaces show a status of up/up.

```
Router#show ip interface brief
```

Step 8: Test Inter-VLAN Connectivity

- From PC0 (VLAN 10), ping PC2 (VLAN 20) and confirm that the reply is received, showing that traffic is now routed between the VLANs through the router's sub-interfaces.

```
ping 192.168.20.10
```

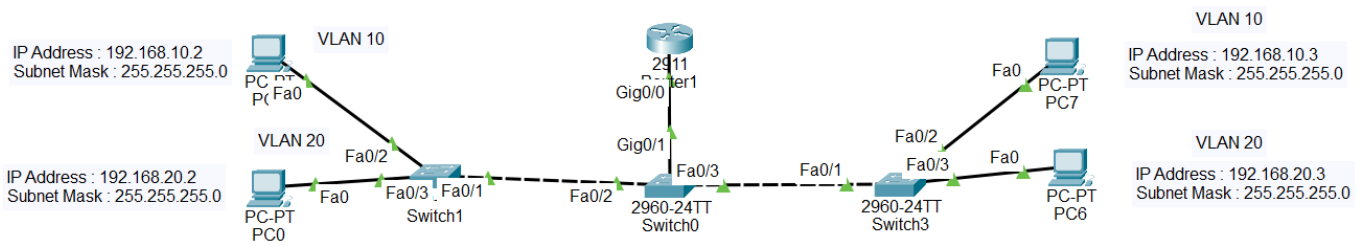
Configuration Table

IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

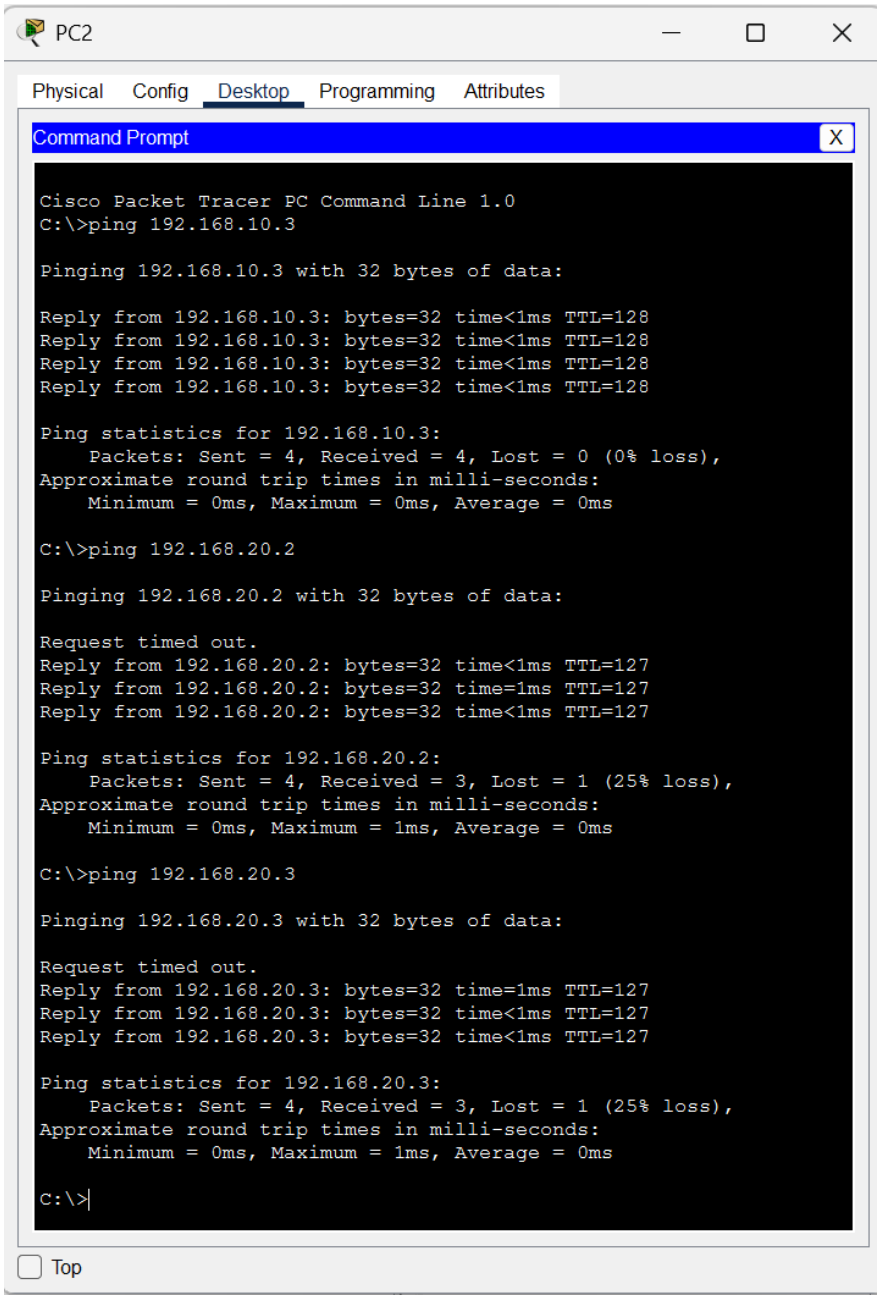
Device	**Interface**	**IP Address**	**Subnet Mask**	**Default Gateway**
Router0	Gi0/0.10 (VLAN 10 gateway)	—	255.255.255.0	—
Router0	Gi0/0.20 (VLAN 20 gateway)	—	255.255.255.0	—
Router0	Gi0/0 (physical, trunk)	Not applicable	—	—

Device	Interface	IP Address	Subnet Mask	Default Gateway
Switch1	Fa0/1 — Trunk to Sw0	Not applicable	—	—
PC0	Fa0/1 — VLAN 10	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Fa0/2 — VLAN 10	192.168.10.3	255.255.255.0	192.168.10.1
PC2	Fa0/3 — VLAN 20	192.168.20.2	255.255.255.0	192.168.20.1
Switch2	Fa0/2 — Trunk to Sw0	Not applicable	—	—
PC3	Fa0/4 — VLAN 20	192.168.20.3	255.255.255.0	192.168.20.1

Circuit Diagram



Output :



Result

Thus, inter-VLAN communication was successfully configured using the router-on-a-stick method in Cisco Packet Tracer. VLANs and the trunk link were configured on the switch through GUI-based configuration, the router's sub-interfaces were configured through the Router CLI using 802.1Q encapsulation, and devices in different VLANs were able to communicate, confirming correct inter-VLAN routing.

Ex no : 11 DHCP Configuration for Automatic IP Allocation

Date :

Aim

To configure a DHCP server in Cisco Packet Tracer so that PCs on a network can automatically obtain IP addressing information.

Procedure

Step 1: Build the Topology

- Drag and drop 1 Switch (2960), 1 Server (Server-PT), and 3 PCs (PC0–PC2).
- Connect the server and all PCs to the switch using Copper Straight-Through cables.

Step 2: Assign a Static IP Address to the Server

- Click the server, open Desktop → IP Configuration, and assign the static IP address listed in the Configuration Table below, since the server itself must not depend on DHCP.

Step 3: Enable the DHCP Service

- On the server, go to the Services tab and select DHCP.
- Turn the service On.

Step 4: Configure the DHCP Pool

- Set the pool parameters — Pool Name LAN_POOL, Default Gateway 192.168.10.1, DNS Server 192.168.10.2, Starting IP Address 192.168.10.3, Subnet Mask 255.255.255.0, Maximum Number of Users 50 — and click Save, then Add.

Step 5: Configure the PCs to Use DHCP

- On each PC, open Desktop → IP Configuration and select the DHCP option instead of Static.
- Observe the PC requesting and receiving an IP address, subnet mask, gateway, and DNS server automatically.

Step 6: Verify the Assigned Addresses

- Open the Command Prompt on each PC and confirm the leased IP configuration.

```
ipconfig /all
```

Step 7: Test Connectivity

- Ping between the PCs and to the server to confirm that DHCP-assigned addressing provides full connectivity.

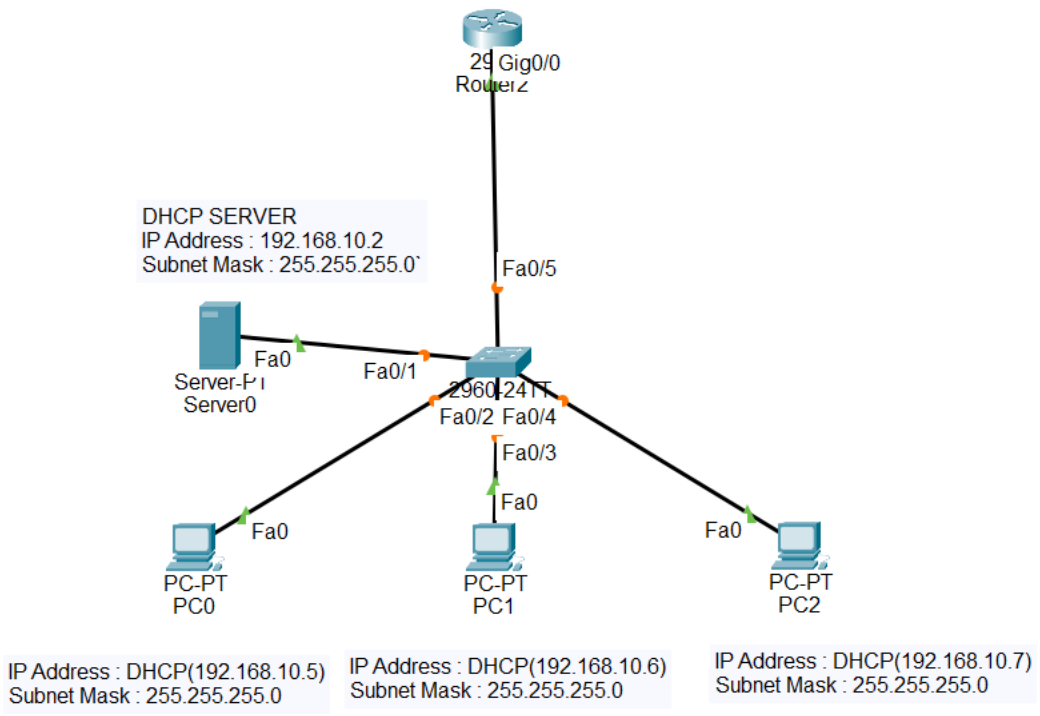
```
ping 192.168.10.2
```

Configuration Table

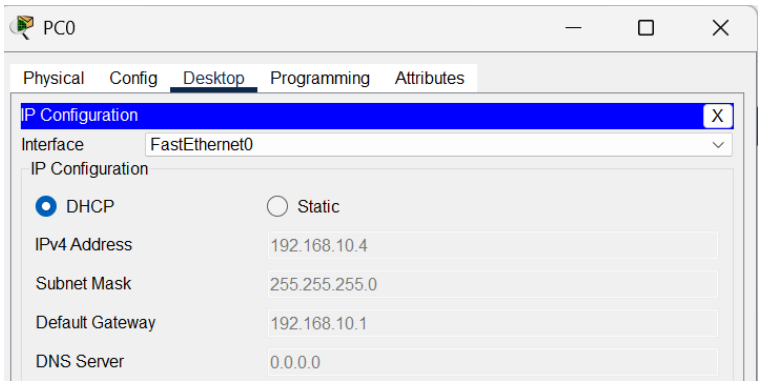
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

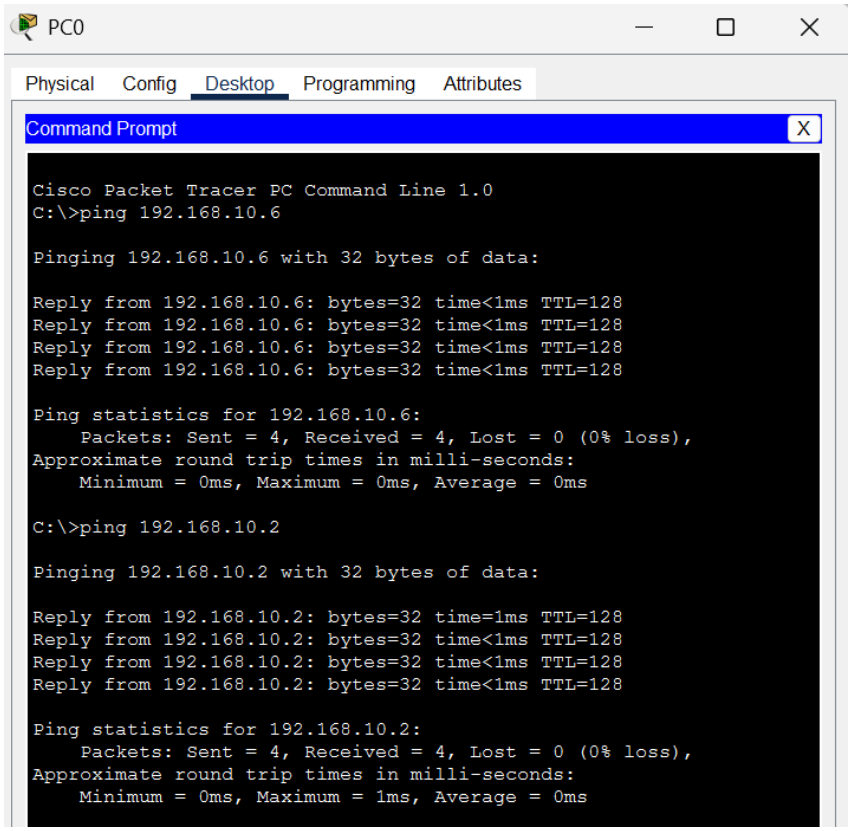
Device	Interface	IP Address	Subnet Mask	Default Gateway
Server0	NIC (static)	192.168.10.2	255.255.255.0	192.168.10.1
PC0	NIC (DHCP)	192.168.10.5(leased)	255.255.255.0	192.168.10.1
PC1	NIC (DHCP)	192.168.10.6(leased)	255.255.255.0	192.168.10.1
PC2	NIC (DHCP)	192.168.10.7(leased)	255.255.255.0	192.168.10.1

Circuit Diagram



Output :





Result

Thus, a DHCP server was successfully configured in Cisco Packet Tracer. PCs on the network automatically obtained valid IP configuration from the server, and connectivity using the leased addresses was verified.

Ex no : 12 DNS Configuration and Name Resolution

Date :

Aim

To configure a DNS server in Cisco Packet Tracer that resolves a domain name to an IP address, and to verify name resolution from a client PC.

Procedure

Step 1: Build the Topology

- Drag and drop 1 Router, 1 Switch , 1 Server (Server-PT) acting as both the DNS and HTTP server, and 3 PC (PC0-PC2).
- Connect the server and PC to the switch using Copper Straight-Through cables.

Step 2: Configure IP Addresses

- Configure the server and PC0 with the addresses listed in the Configuration Table below.

Step 3: Enable the DNS Service

- On the server, go to the Services tab and select DNS.
- Turn the service On.

Step 4: Add a DNS Record

- Enter Type: A Record, Name: www.college.com, Address: 192.168.10.200, and click Add to create the record.

Step 5: Enable the HTTP Service on the Server

- Go to the Services tab, select HTTP, and confirm that the service is On so the domain name resolves to a working web server.

Step 6: Configure the DNS Setting on PC

- Open Desktop → IP Configuration on PCs and set the DNS Server field to 192.168.10.2.

Step 7: Verify Name Resolution

- Open a web browser on PC0 and enter the domain name instead of the IP address.
- Confirm that the server's default web page loads, proving that the domain name was correctly resolved to its IP address.

  http://www.college.com

Step 8: Verify Using the Command Line

- Optionally, use the ping command with the domain name and observe that it resolves to the configured IP address before sending Echo Requests.

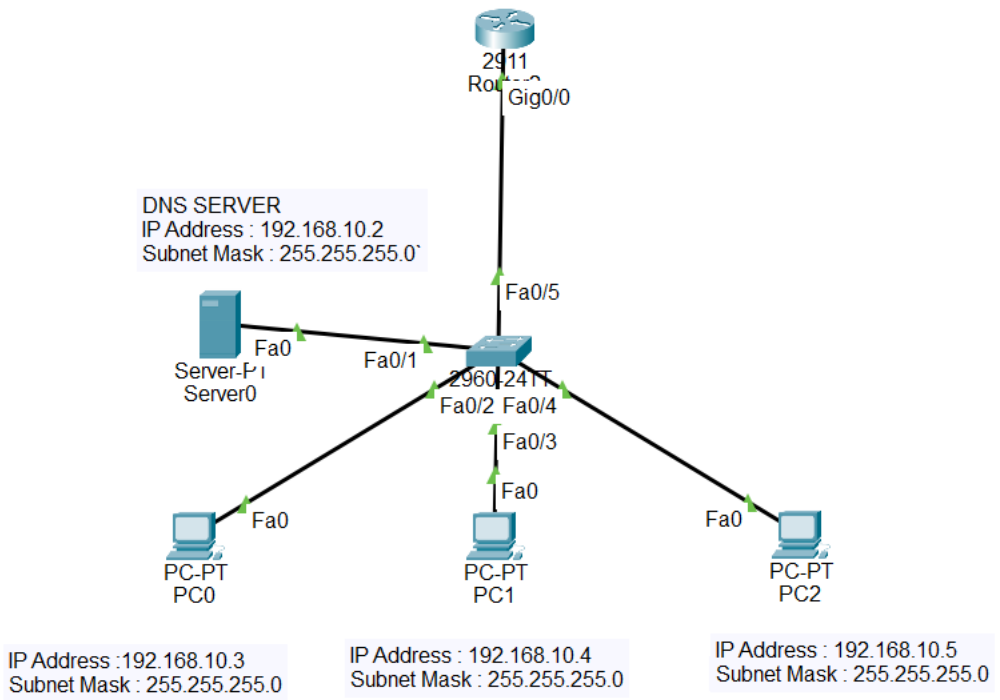
  ping www.college.com

Configuration Table

IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	Interface	IP Address	Subnet Mask	Default Gateway
Server0	NIC (DNS + HTTP)	192.168.10.2	255.255.255.0	192.168.10.1
PC0	NIC (DNS: 192.168.10.2)	192.168.10.3	255.255.255.0	192.168.10.1
PC0	NIC (DNS: 192.168.10.2)	192.168.10.4	255.255.255.0	192.168.10.1
PC0	NIC (DNS: 192.168.10.2)	192.168.10.5	255.255.255.0	192.168.10.1

Circuit Diagram



Output :



Result

Thus, a DNS server was successfully configured in Cisco Packet Tracer. A domain name was mapped to an IP address, and successful name resolution was verified from a client PC using both a web browser and the ping command.

Ex no : 13 HTTP Web Server and Client Communication

Date :

Aim

To configure an HTTP server in Cisco Packet Tracer and verify client-server communication using the HTTP application layer protocol through a web browser.

Procedure

Step 1: Build the Topology

- Drag and drop 1 Switch, 1 Server (Server-PT),1 PCs (PC0) and 1 Laptop.
- Connect the server and both PCs to the switch using Copper Straight-Through cables.

Step 2: Configure IP Addresses

- Configure the server and both PCs with the addresses listed in the Configuration Table below.

Step 3: Enable the HTTP Service

- Click the server, go to the Services tab, and select HTTP.
- Confirm that the HTTP (and HTTPS, if required) service is turned On.

Step 4: Edit the Default Web Page (Optional)

- Within the HTTP service editor, modify the index.html content so the page can be identified as the lab's server during testing.

Step 5: Access the Server from PC0

- Open Desktop → Web Browser on PC0.
- Enter the server's IP address in the URL bar and load the page.

  http://192.168.70.1

Step 6: Access the Server from PC1

- Repeat the process on Laptop0 to confirm that multiple clients can independently request pages from the same HTTP server.

Step 7: Observe HTTP in Simulation Mode

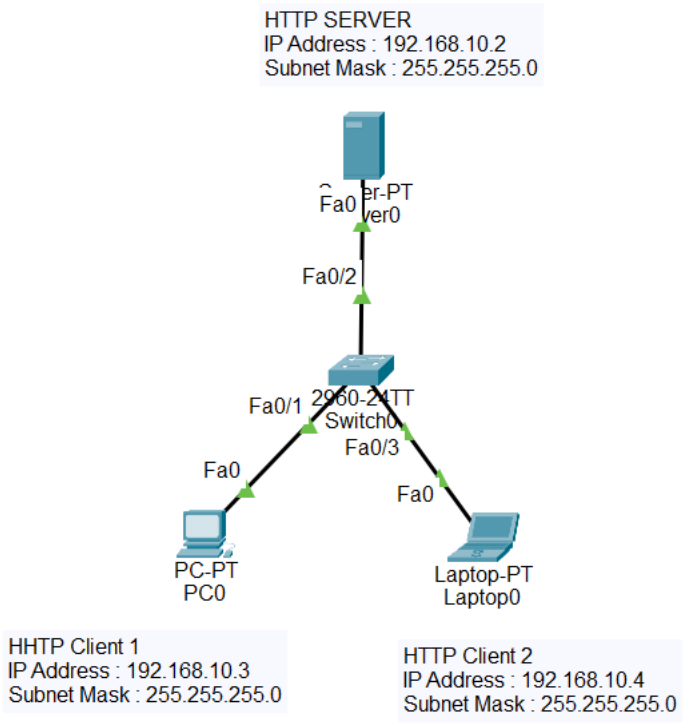
- Switch to Simulation Mode and filter for HTTP and TCP protocols.
- Generate an HTTP request from a PC's web browser and step through the events to observe the TCP three-way handshake followed by the HTTP GET request and response.

Configuration Table

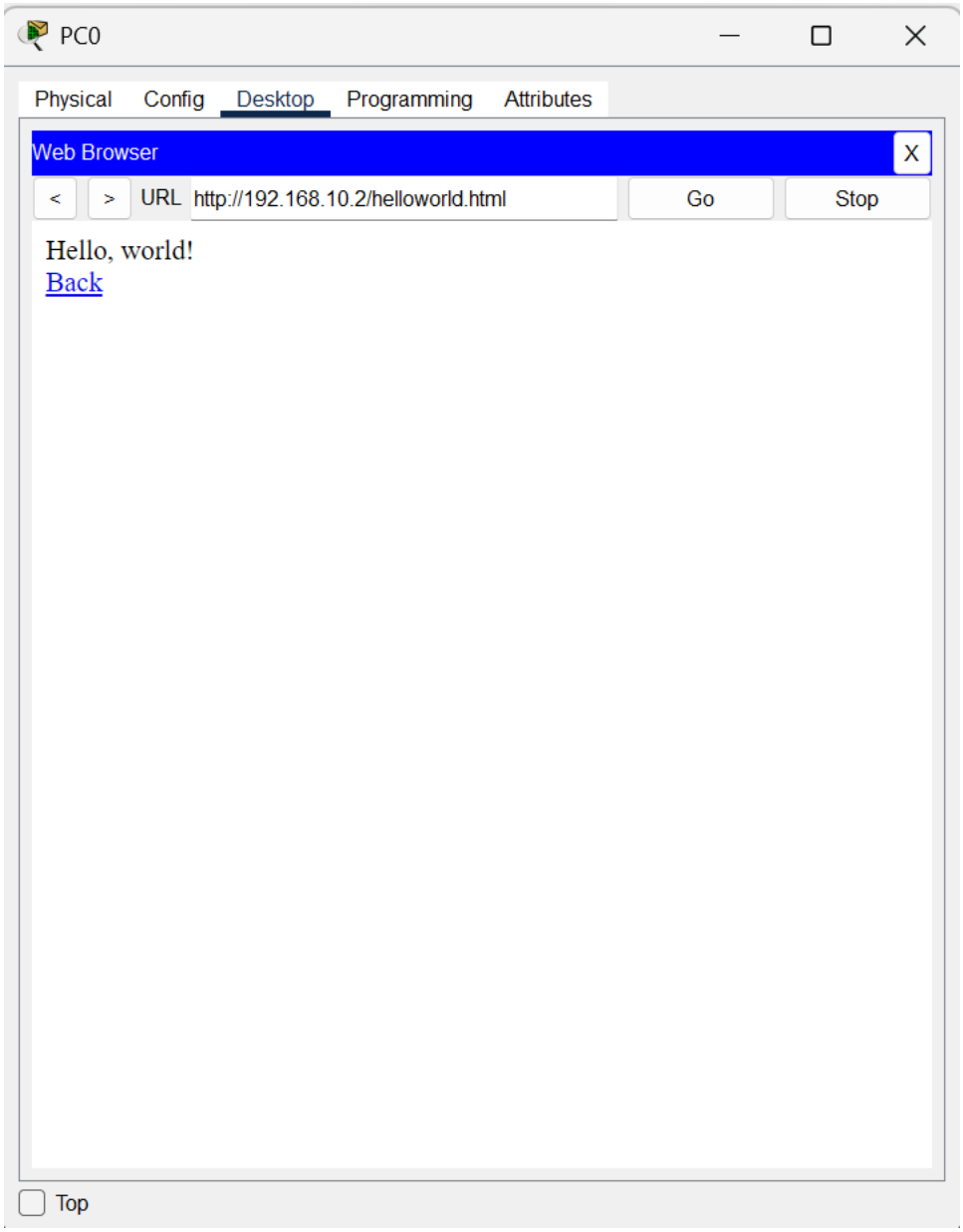
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	Interface	IP Address	Subnet Mask	Default Gateway
Server0	NIC (HTTP)	192.168.10.2	255.255.255.0	—
PC0	NIC	192.168.10.3	255.255.255.0	—
Laptop0	NIC	192.168.10.4	255.255.255.0	—

Circuit Diagram



Output :



Result

Thus, an HTTP server was successfully configured in Cisco Packet Tracer, and client-server communication using the HTTP protocol was verified from multiple PCs. The underlying TCP connection establishment and HTTP request-response exchange were observed in Simulation Mode.

Ex no : 14 FTP File Transfer Between Client and Server

Date :

Aim

To configure an FTP server in Cisco Packet Tracer and verify file transfer between a client and the server using FTP commands.

Procedure

Step 1: Build the Topology

- Drag and drop 1 Switch , 1 Server (Server-PT), 1 PC (PC0) and 1 Laptop(Laptop0).
- Connect the server and PC to the switch using Copper Straight-Through cables.

Step 2: Configure IP Addresses

- Configure the server ,PC0 and Laptop0 with the addresses listed in the Configuration Table below.

Step 3: Enable the FTP Service

- Click the server, go to the Services tab, and select FTP.
- Turn the service On.

Step 4: Create an FTP User Account

- Add username labuser with password lab123, granting Read, Write, and List permissions.

Step 5: Connect to the FTP Server from PC0

- Open the Command Prompt on PC0 and start an FTP session with the server.
- Enter the username and password created in Step 4 when prompted.

  ftp 192.168.10.2

Step 6: Upload a File to the Server

- Use the put command to transfer a file from the PC to the server.

  ftp> put report.txt

Step 7: Download a File from the Server

- Use the get command to transfer a file from the server to the PC.

  ftp> get welcome.txt

Step 8: List Files and Close the Session

- Use the dir command to list files available on the server, then close the connection.

  ftp> dir

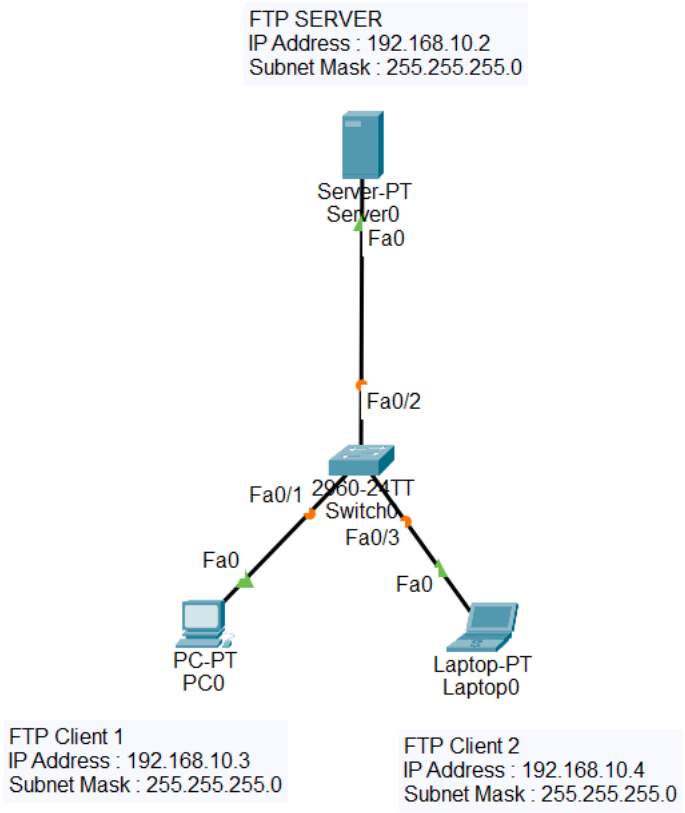
  ftp> quit

Configuration Table

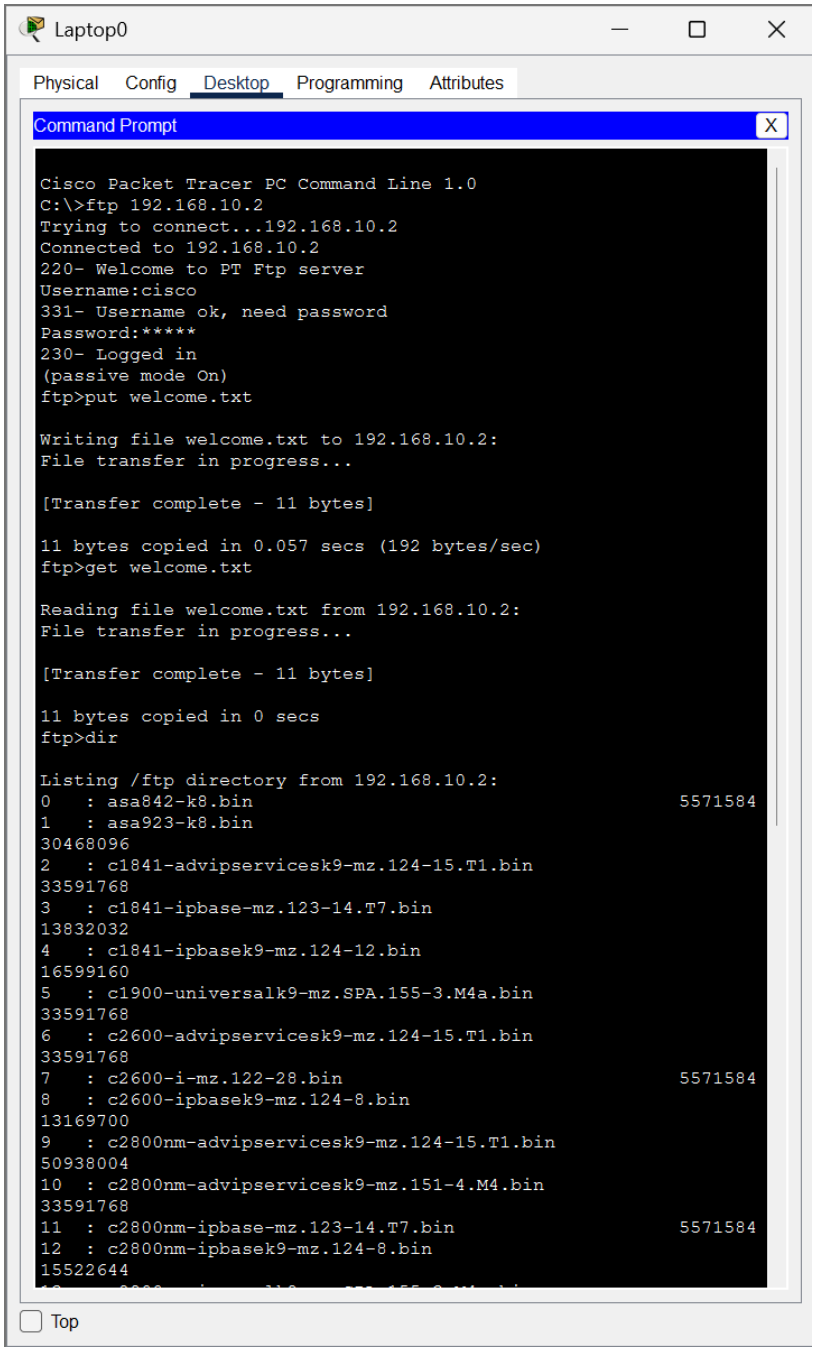
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

Device	Interface	IP Address	Subnet Mask	Default Gateway
Server0	NIC (FTP)	192.168.10.2	255.255.255.0	—
PC0	NIC	192.168.10.3	255.255.255.0	—
Laptop0	NIC	192.168.10.4	255.255.255.0	—

Circuit Diagram



Output :



Result

Thus, an FTP server was successfully configured in Cisco Packet Tracer, and file transfer between the client and server was verified using the put, get, and dir commands over the FTP protocol.

Ex no : 15 TCP vs UDP Basic Communication Simulation

Date :

Aim

To simulate and compare TCP and UDP communication in Cisco Packet Tracer, and to observe the differences in connection establishment and data delivery between the two transport layer protocols.

Procedure

Step 1: Build the Topology

- Drag and drop 1 Switch , 1 Server (Server-PT), and 2 PCs (PC0,PC1).
- Connect the server and PC to the switch using Copper Straight-Through cables.

Step 2: Configure IP Addresses

- Configure the server and PC0 with the addresses listed in the Configuration Table below.

Step 3: Enable TCP- and UDP-Based Services

- On the server, ensure the HTTP service (which uses TCP) is turned On.
- Ensure the TFTP service (which uses UDP) is turned On, so both protocol types are available to test.

Step 4: Switch to Simulation Mode

- Click Simulation Mode and open Edit Filters.
- Select TCP and HTTP only, deselecting other protocols to keep the event list readable.

Step 5: Generate and Observe TCP Traffic

- From PC0's web browser, request the server's default page.
- Step through the event list and observe the three-way handshake: SYN, SYN-ACK, and ACK segments, exchanged before any HTTP data is sent.
- Note the sequence and acknowledgement numbers shown in the PDU details, and observe the connection teardown (FIN/ACK) at the end.

Step 6: Reset Filters for UDP Traffic

- Clear the event list and update Edit Filters to select UDP and TFTP only.

Step 7: Generate and Observe UDP Traffic

- From PC0's Command Prompt, initiate a TFTP request to the server.
- Step through the event list and observe that data segments are sent directly, with no handshake or acknowledgement sequence preceding the transfer.

  tftp

Step 8: Compare and Record the Observations

- Tabulate the observed differences between the two protocols: connection setup, reliability, overhead, and typical use.
- TCP (HTTP) uses a three-way handshake (SYN, SYN-ACK, ACK), is acknowledged and retransmits lost segments, carries higher overhead, and suits web browsing and file transfer requiring reliability.

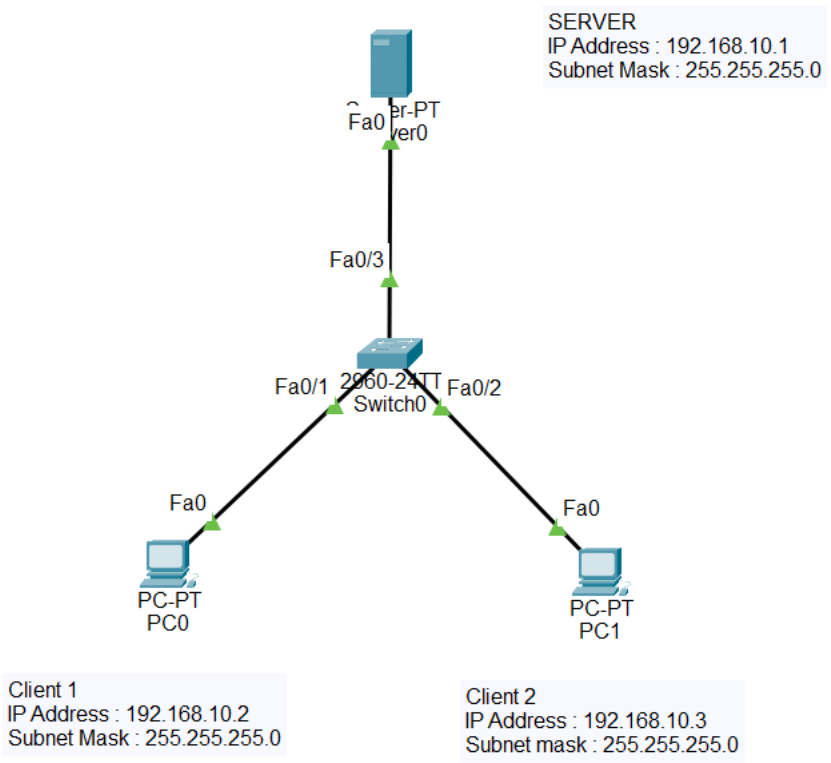
- UDP (TFTP) uses no handshake, is best-effort with no acknowledgement by default, carries lower/minimal header overhead, and suits streaming and simple transfers tolerant of loss.

Configuration Table

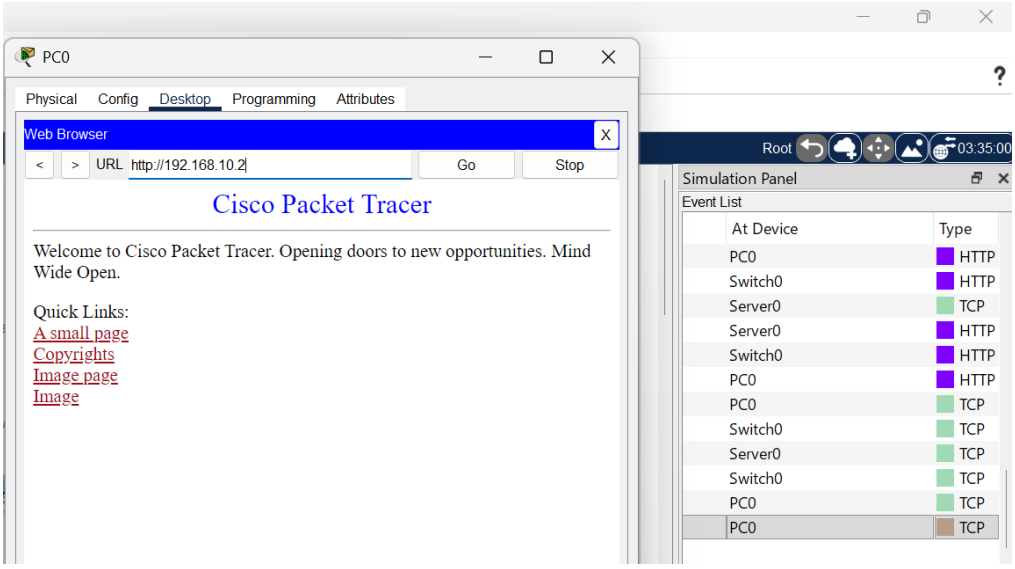
IP addressing details for every device used in this exercise (PCs, routers, switches, servers, and other end devices):

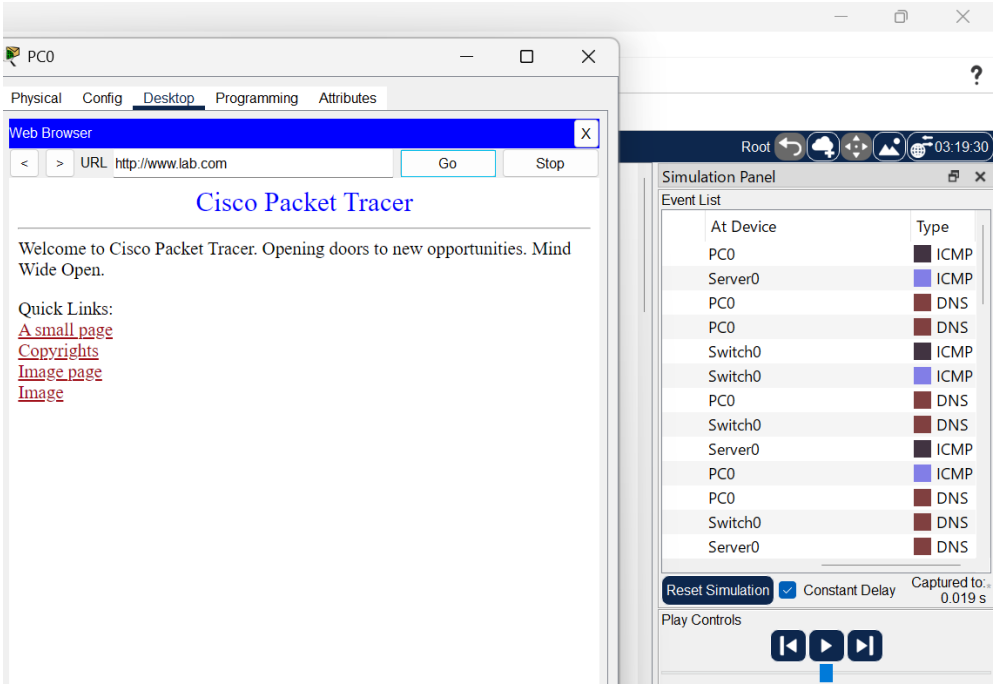
Device	Interface	IP Address	Subnet Mask	Default Gateway
Server0	—	192.168.10.1	255.255.255.0	—
PC0	NIC	192.168.10.2	255.255.255.0	—
PC1	NIC	192.168.10.3	255.255.255.0	—

Circuit Diagram



Output :





Result

Thus, TCP and UDP communication were successfully simulated in Cisco Packet Tracer. The connection-oriented, handshake-based behaviour of TCP was compared with the connectionless behaviour of UDP, and the key differences between the two transport layer protocols were observed and recorded.